

# A binomial count of meromorphic travelling waves

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## Abstract

We study meromorphic travelling waves of  $u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0$ , a constant-coefficient family containing KdV–Burgers, Kuramoto–Sivashinsky and Kawahara. After integration and normalization, the profiles satisfy  $P(D)v + v^2/2 = 0$ , where  $D = d/dz$  and  $P$  is monic of degree  $p \geq 2$  with complex coefficients.

For evolution equations of even order ( $p$  odd), and for purely dispersive equations (only odd-order derivatives), every nonconstant meromorphic travelling wave is rational, rational in an exponential, or elliptic, with one pole per period.

One-pole waves of these three kinds are exceptional: for  $p \geq 3$  a generic equation of the family has none. We count the exceptions as pairs  $(P, v)$ , over the whole family: there are  $\binom{2p-1}{p-1}$  with  $v$  rational in an exponential of given rate, and  $2\binom{2p-2}{p-2}$  with  $v$  elliptic on a given generic lattice. We also count the solitary pulses of the purely dispersive subfamily and the fronts of the purely dissipative one (only even-order derivatives in the linear part).

When  $2p-1$  is prime, these statements are proved and all pairs are distinct. We conjecture them in every order.

**Keywords.** Meromorphic travelling waves; dispersive and dissipative equations; elliptic functions; polynomial convolution; algebraic multiplicity.

## 1 Introduction

Exact travelling waves of nonlinear evolution equations are usually found one equation at a time, and usually only for special values of the coefficients. The Kuramoto–Sivashinsky equation with dispersion,  $u_t + uu_x + u_{xx} + \sigma u_{xxx} + u_{xxxx} = 0$ , is a classical example. It has nonconstant travelling waves that are polynomials in a hyperbolic tangent for exactly four values of the dispersion coefficient, up to its sign:  $\sigma = 0, 4, 12/\sqrt{47}$  and  $16/\sqrt{73}$ . Kuramoto and Tsuzuki found the wave of the dispersionless equation [62], Kudryashov the others [32–35], and Eremenko proved that, together with Kudryashov’s elliptic solution and a single rational one, they exhaust the nonconstant meromorphic travelling waves [23]. The classical tables list six cases at these four values [7, 43]: two at  $\sigma = 0$ , two at  $\sigma = \pm 4$ , and one at each of the other two values. The four values already appear together in Kudryashov [33, Eq. (39)]. The tables do not distinguish a wave from its mirror image, which solves the equation with the opposite sign of  $\sigma$ . If the two are counted separately, the two cases with  $\sigma = 0$  are their own mirror images and the other four are not, so the list has  $2 + 2 \cdot 4 = 10$  members. This paper shows that ten is  $\binom{5}{2}$ : the classical list is the case  $p = 3$  of a binomial enumeration, proved for infinitely many  $p$  and conjectured for all.

We consider the evolution equations

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0,$$

which contain the KdV–Burgers ( $p = 2$ ), Kuramoto–Sivashinsky ( $p = 3$ ), Kawahara and Benney–Lin ( $p = 4$ ) and Nikolaevskiy ( $p = 5$ ) equations; Table 1 lists the named members with  $p \leq 6$ . Kudryashov studied this family as a whole and found exact waves of its KdV–Burgers, Kuramoto–Sivashinsky and Kawahara members [33]. A travelling wave  $u = w(x - ct)$  satisfies, after one integration,  $\sum_j b_j w^{(j)} - cw + w^2/2 + K = 0$ . Shifting  $w$  by a constant solution and normalizing the leading coefficient gives

$$P(D)v + \frac{1}{2}v^2 = 0, \quad P(\rho) = \rho^p + \sum_{j=0}^{p-1} a_j \rho^j, \quad p \geq 2, \quad (1.1)$$

with  $D = d/dz$  and complex coefficients. The coefficients  $a_1, \dots, a_{p-1}$  are those of the evolution equation, up to a rescaling of  $x$ , and the speed, the integration constant and the background are absorbed into  $a_0$  (Section 2). Throughout,  $p$  is the order of the profile equation (1.1), which is one less than the order of the evolution equation: Kuramoto–Sivashinsky is a fourth-order equation and has  $p = 3$ . Phrases such as “order three” or “in every order” always refer to  $p$ .

In the evolution equation, derivatives of odd order are dispersive and those of even order dissipative (in this sign convention a positive coefficient of  $u_{xx}$  is gain, so dissipative includes gain). An equation is *purely dispersive* if it contains only odd-order derivatives, as KdV and Kawahara do; then  $p$  is even and  $P$  is an even polynomial. It is *purely dissipative* if its linear part contains only even-order derivatives, as for Kuramoto–Sivashinsky without dispersion; then  $p$  is odd and  $P - P(0)$  is odd. These are *the two pure subfamilies*, one for each parity of  $p$ . All other equations, such as KdV–Burgers, are *mixed*.

**The waves.** A travelling wave is *meromorphic* if its profile is a single-valued meromorphic function of the complex variable  $z$ . The familiar exact solutions are of this kind: a polynomial in  $\tanh$  or  $\operatorname{sech}$  is a rational function of an exponential  $e^{\kappa z}$ , a cnoidal wave is an elliptic function, and rational functions of  $z$  also occur. These are the three kinds of wave in this paper. We consider those with *one pole per period*: all the poles of the profile are translates of a single pole by its periods. For a wave of  $\tanh$  type the period is the imaginary period  $2\pi i/\kappa$  of  $e^{\kappa z}$ , and the rate  $\kappa$  measures the width of the wave. Such a wave tends to a constant at each end (Section 2.3); it is a *pulse* if it returns to the same constant, and a *front* if it connects two different ones.

For every equation of even order, and for every purely dispersive equation, these are all the meromorphic travelling waves that exist (Theorem 5.1). For the remaining equations, the mixed ones of odd order such as KdV–Burgers, other meromorphic waves can occur; there we count the one-pole waves without claiming more. These equations are the rows marked  $\dagger$  in Table 1.

**The question.** For  $p \geq 3$  such waves are exceptional: an equation of the family has one only if its coefficients satisfy algebraic relations, as the four values of  $\sigma$  illustrate. This suggests the right question. We do not ask how many waves a given equation has, but how many pairs of an equation and a wave there are in the whole family.

To make this question finite, two continuous freedoms must be removed. A translate of a wave is again a wave, so we place the pole at a fixed point. Rescaling  $x$  changes the width of a wave and the coefficients of its equation together, so we fix the width: for a wave that is rational in an exponential we prescribe the rate  $\kappa = 1$ , and for an elliptic wave we prescribe both of its periods, that is, its period lattice. The coefficients of the equation remain free. We therefore count pairs  $(P, v)$ , where  $P$  ranges over all monic polynomials of degree  $p$  and  $v$  solves (1.1):

- a *rational-exponential pair* has  $v$  rational in  $e^z$ , with one pole modulo  $2\pi i\mathbb{Z}$ , placed at  $i\pi$ , so that the profile is smooth on the real axis, and  $v \rightarrow 0$  as  $e^z \rightarrow 0$ , that is, as  $z \rightarrow -\infty$  along the real axis;
- an *elliptic pair* has  $v$  elliptic on a given lattice, with its only pole at zero.

A wave read in the opposite direction,  $z \mapsto -z$ , is counted separately. At the same rate it is a wave of the mirror-image equation, in which the terms of one parity have changed sign, and restoring the normalization gives again a pair, the *mirror image* of the first. For a fixed equation it is the same wave, read with the rate  $-\kappa$ .

One more convention explains a factor two below. The profile equation has two constant solutions, and a wave can be measured from either of them. The change is a Galilean shift: it changes the speed relative to the background, and with it  $P$ , but not the wave (Section 2). A rational-exponential wave is measured from its limit as  $e^z \rightarrow 0$ . An elliptic wave has no such limit, and both choices are counted, so every elliptic wave appears in two pairs.

**The answer.** Four numbers appear:

$$\begin{aligned}
N_p &= \binom{2p-1}{p-1} && \text{rational-exponential pairs,} \\
S_p &= \binom{2p-2}{p-2} && \text{the pulses among them; the other } N_p - S_p \text{ are fronts,} \\
M_p &= 2S_p && \text{elliptic pairs on a generic lattice,} \\
F_p &= \binom{p-1}{\lfloor (p-1)/2 \rfloor} && \text{rational-exponential pairs of the pure subfamily of order } p.
\end{aligned}$$

We call  $2p-1$  the *index* of the order  $p$ . Our main result establishes these numbers whenever the index is a prime number:

- there are exactly  $N_p$  rational-exponential pairs, and they are all distinct (Theorem 3.4);
- exactly  $S_p$  of them are pulses, and the others are fronts (Proposition 3.6);
- a generic lattice carries exactly  $M_p = 2S_p$  elliptic pairs, and they are all distinct (Theorem 4.1); generic means every lattice except those with finitely many values of the invariant  $j(\Lambda)$ , which at  $p=3$  are  $j=0$  and  $-300$ ;
- exactly  $F_p$  of the rational-exponential pairs belong to the pure subfamily of order  $p$ , and they are the pairs that coincide with their mirror image. Those of a purely dispersive equation are pulses, and those of a purely dissipative one are fronts. For even  $p$ ,  $2F_p$  of the elliptic pairs belong to purely dispersive equations (Theorem 6.2, Corollary 6.4).

The index is prime for  $p=2, 3, 4, 6, 7, 9, 10, 12, \dots$ . At the three remaining orders up to twelve,  $p=5, 8$  and  $11$ , the same statements are established by exact computer calculations (Corollary 9.2, Proposition 9.1(iii) and Appendix A). We call the orders with a prime index, together with all  $p \leq 12$ , the *covered* orders; they include every named equation of Table 1. We conjecture the same statements in every order (Conjecture 9.3). At a prime index the first statement has also been verified formally, in Lean (Remark 3.5).

The names and some physical sources: Benney–Lin [4, 63]; Kuramoto–Sivashinsky with dispersion, also called the generalized Kuramoto–Sivashinsky, Benney or KdV–KS equation [29];

Table 1: The named equations with  $p \leq 6$ ;  $u_{kx} = \partial_x^k u$ . Each name stands for a family with free coefficients; the numbers count distinct pairs across it, at unit rate and on a generic lattice. A dagger marks the families that may have meromorphic travelling waves of other kinds; the others have none, apart from a rational wave when all lower coefficients vanish.

| Equation                     | Linear terms             | $p$ | Rational-exponential pairs | Elliptic pairs |
|------------------------------|--------------------------|-----|----------------------------|----------------|
| KdV                          | $u_{3x}$                 | 2   | $F_2 = 1$                  | $2F_2 = 2$     |
| KdV–Burgers <sup>†</sup>     | $u_{2x}, u_{3x}$         | 2   | $N_2 = 3$                  | $M_2 = 2$      |
| Kuramoto–Sivashinsky (KS)    | $u_{2x}, u_{4x}$         | 3   | $F_3 = 2$                  | 0              |
| KS with dispersion           | $u_{2x}, u_{3x}, u_{4x}$ | 3   | $N_3 = 10$                 | $M_3 = 8$      |
| Kawahara                     | $u_{3x}, u_{5x}$         | 4   | $F_4 = 3$                  | $2F_4 = 6$     |
| Benney–Lin <sup>†</sup>      | $u_{2x}, \dots, u_{5x}$  | 4   | $N_4 = 35$                 | $M_4 = 30$     |
| Nikolaevskiy                 | $u_{2x}, u_{4x}, u_{6x}$ | 5   | $F_5 = 6$                  | 0              |
| Nikolaevskiy with dispersion | $u_{2x}, \dots, u_{6x}$  | 5   | $N_5 = 126$                | $M_5 = 112$    |
| seventh-order KdV            | $u_{3x}, u_{5x}, u_{7x}$ | 6   | $F_6 = 10$                 | $2F_6 = 20$    |

Nikolaevskiy [69], with dispersion [41, 72]; long waves in a liquid with gas bubbles [52, 53] and in a viscoelastic tube [40, 44], where the mixed fifth-order equation of the Benney–Lin row was also derived [54] and studied numerically [55]. The rows with  $N_p$  and  $M_p$  are full families, and a full family contains the pure subfamily of the same  $p$ . The small entries are familiar. The three pairs of KdV–Burgers are the KdV soliton and the KdV–Burgers front, read in its two directions. The two elliptic pairs of KdV are the cnoidal wave on its two backgrounds. The eight elliptic pairs at  $p = 3$  are Kudryashov’s elliptic solution: on a given lattice it occurs four times, always with  $\sigma = \pm 4$ , and each time on two backgrounds (Section 8.2). The two entries 0 say that these purely dissipative equations have no elliptic pair on a generic lattice (Section 6); at  $p = 3$  they have one only when  $g_2 = 0$ . The six pairs of the purely dissipative family with  $p = 5$  are the six solitary waves of Kudryashov and Demina [49], four of them real [41] (Section 8.4).

The counts are over  $\mathbb{C}$ , but many pairs have real coefficients and give smooth bounded waves of real equations: eight of the ten at  $p = 3$  and 21 of the 35 at  $p = 4$  (Table 2, Section 8.3). A pair with non-real coefficients at unit rate can still belong to a real equation, at an imaginary rate; the wave is then periodic in  $x$  and has poles on the real axis (Section 3.2). *Distinct* carries content: like a root of a polynomial, a solution of a polynomial system can be multiple, and on special lattices elliptic pairs do merge in this way (Section 8.2). Rational waves are not counted: at the covered orders the only one with a single pole is  $v = c_* z^{-p}$ , for  $P = D^p$  [45], with  $c_*$  as in (2.7). The general enumeration has no precedent that we know of. Its special cases at  $p = 3$ , in the Kawahara family [19, 33] and in the Nikolaevskiy family [49] are in the literature (Sections 3.2, 8). What is new is a uniform explanation across orders of why the lists are complete and how long they are. We do not discuss stability, and treat only the quadratic nonlinearity  $uu_x$ .

**A single equation.** The numbers in Table 1 are totals over all coefficient values of a family. Rescaling  $x$  moves an equation along a curve in the space of coefficient ratios, and each pair marks one curve (Section 7). The equations with a one-pole rational-exponential wave are those on these curves; whenever  $2p - 1$  is prime, and for every  $p \leq 573$ , the curves are finite in number (Corollary 7.2). This is the precise sense in which the waves are exceptional for  $p \geq 3$ ; at  $p = 3$  the curves are  $b_2^2 = \sigma^2 b_1 b_3$  with the four values of  $\sigma$ . In a family with a single ratio (KdV–Burgers,

Kawahara, Kuramoto–Sivashinsky without dispersion) a possibly complex rescaling carries each equation with a nonzero ratio to any other, so each has the KdV–Burgers front, respectively all  $F_4 = 3$  or  $F_3 = 2$  wave shapes of its family. An equation without lower-order terms, such as KdV, has each of its waves at every rate.

For equations of even order the classification gives more: at the same orders a generic such equation (outside a proper algebraic subset of the coefficients) has no nonconstant meromorphic travelling wave at all (Corollary 7.3; Remark 7.4 for  $p = 3$ ). It may well have travelling waves; but none of them is meromorphic. The counts also apply to the many other equations, with mixed derivatives or in several space variables, that have the same profile equation. They are a checklist for the output of expansions in powers of  $\tanh$  or a Jacobi function: always for the equations of Theorem 5.1, otherwise for outputs with one pole per period.

**When these are all the meromorphic waves.** Theorem 5.1 is the classification stated above, for equations of even order and purely dispersive ones. The proof rests on Eremenko’s theorem [23], which requires finitely many Laurent expansions at a pole. For odd  $p$  the expansion is unique. For a purely dispersive equation one Laurent coefficient is free, a resonance; but the equation conserves an energy, and fixing the energy fixes that coefficient. By the same uniqueness, every wave of a pure subfamily is symmetric about its poles, or antisymmetric up to a constant (Corollary 5.3). The theorem is a uniform version of results that exist in pieces (Section 5): odd  $p$  is contained in Yuan et al. [75, Theorem 1], and the energy device is that of Eremenko et al. [24, Lemmas 3–4]. The polynomial systems for the waves are instances of the principal-part construction of Demina and Kudryashov [20, 21]. The later work of Demina [16, 18] develops Nevanlinna and algebraic-invariant methods for meromorphic classification, including equations without finitely many Laurent expansions. A systematic account appears in Belova [3, Chapter 5]. In particular, her Theorem 5.8 classifies the elliptic and rational-exponential solutions of a mixed fourth-order quadratic equation and proves their one-pole property. This fixed-equation classification is distinct from the present count of operator–wave pairs with prescribed geometry. Mixed equations of odd order cannot be included: the profile equation of KdV–Burgers is also that of the Fisher equation, and at the speed of Ablowitz and Zeppetella it has meromorphic solutions of none of these kinds (Remark 5.2).

**Every wave has a label.** Here is where the binomial coefficient  $N_p$  comes from. Put  $Q = 1/(1 + e^{-z}) = (1 + \tanh(z/2))/2$ . Every rational-exponential pair has a profile of the form  $v_A = s_p A(D)Q$ , where  $s_p = 2(2p - 1)!/((p - 1)!)^2$  is a fixed scale ( $s_2 = 12$ ,  $s_3 = 60$ ) and  $A$  is a monic polynomial of degree  $p - 1$ ; this is a polynomial in  $\tanh(z/2)$  written in a convenient basis. For  $\operatorname{Re} z < 0$  we have  $Q = e^z - e^{2z} + e^{3z} - \dots$ , and  $D$  multiplies  $e^{nz}$  by  $n$ . Hence the coefficient of  $e^{nz}$  is  $\pm s_p A(n)$  in  $v_A$  and  $\pm s_p P(n)A(n)$  in  $P(D)v_A$ , while in  $v_A^2$  it is  $\pm s_p^2 S_A(n)$  with the convolution sum

$$S_A(n) = \sum_{j=1}^{n-1} A(j)A(n-j).$$

These sums are the values of a polynomial  $S_A$  of degree  $2p - 1$ , and (1.1) holds exactly when  $2P(n)A(n) = s_p S_A(n)$  for all  $n$ . So the pair exists if and only if the polynomial  $A$  divides the polynomial  $S_A$ , and then  $P$  is the quotient, up to the constant factor (Proposition 3.1).

The smallest case shows what happens. For  $p = 2$  and  $A = \rho + a$ ,

$$S_A(\rho) = (\rho - 1) \left( \frac{\rho(\rho + 1)}{6} + a\rho + a^2 \right),$$

which is divisible by  $\rho + a$  exactly when  $a(a - 1)(a + 1) = 0$ . The solution  $a = 0$  is the KdV soliton, and  $a = \pm 1$  are the KdV–Burgers front read in its two directions. Now reduce modulo  $3 = 2p - 1$ . Whatever the value of  $a$ ,

$$3S_A(\rho) \equiv -(\rho^3 - \rho) = -\rho(\rho - 1)(\rho + 1) \pmod{3},$$

so modulo 3 the divisor  $A$  must be one of the three factors  $\rho$ ,  $\rho - 1$ ,  $\rho + 1$ . Each of them comes from exactly one solution, and the root of  $A$  modulo 3 serves as a label:  $\{0\}$  for the soliton,  $\{2\}$  for the front  $a = 1$ , that is, the root  $-1 \equiv 2$ , and  $\{1\}$  for the front  $a = -1$ .

The same happens whenever  $\ell = 2p - 1$  is prime. The polynomial  $S_A$  is a combination of the binomial coefficients  $\binom{\rho-1}{k}$ , and only the last of them, with  $k = \ell$ , has  $\ell$  in its denominator. For this reason  $\ell S_A$  reduces modulo  $\ell$  to  $\pm(\rho^\ell - \rho)$ , whatever  $A$  is, and the condition becomes that  $A$  divides  $\rho^\ell - \rho$  modulo  $\ell$ . Now  $\rho^\ell - \rho$  is the product of  $\rho - s$  over all residues  $s$  modulo  $\ell$ . Its monic divisors of degree  $p - 1$  are therefore given by the  $(p - 1)$ -element subsets  $L$  of the integers modulo  $\ell$ , and there are  $\binom{\ell}{p-1} = N_p$  of them. We prove that every pair can be reduced modulo  $\ell$ , although its coefficients are in general algebraic numbers and not integers, and that every divisor comes from exactly one pair. An algebraic number is reduced through an equation that it satisfies:  $i$  is 2 or 3 modulo 5, because  $2^2 \equiv 3^2 \equiv -1$ . For instance  $A = \rho^2 + 1/11$ , the Kuramoto–Tsuzuki wave, reduces to  $\rho^2 + 1 = (\rho - 2)(\rho - 3)$  modulo 5, and has the label  $\{2, 3\}$ .

So each pair has a label  $L$ , and the labels carry information. The pair is a pulse exactly when  $L$  contains 0. At  $p = 3$  the ten classical waves correspond to the ten two-element subsets of  $\{0, 1, 2, 3, 4\}$  (Table 2), and the four pulses are the subsets that contain 0. The mirror image of the pair with label  $L$  has label  $-L$ . The pairs with  $L = -L$  coincide with their mirror image, and they are exactly the pairs of the two pure subfamilies. There are  $F_p$  such subsets.

The elliptic count comes from the pulses. When one period of the lattice tends to infinity, an elliptic pair becomes a pulse, measured from one of its two constant backgrounds. We show that every pulse, on either background, arises in this way exactly once, and this gives  $M_p = 2S_p$ .

**Other orders.** The number  $N_p$  is also the largest number of isolated solutions, counted with multiplicity, that matching equations of this shape can have (Proposition 3.2). It is attained if no solution lies at infinity, which is rigidity, and the solutions are distinct if all of them are simple. These are the two facts that are not yet proved at every order. For waves, rigidity says that  $c_* z^{-p}$  is the only rational wave with a single pole (Lemma 3.3). It is proved when  $2p - 1$  is a prime or a prime power and established by exact computations for  $p \leq 573$ ; simplicity is known only at the covered orders (Section 9 and a companion note).

**Organization.** The counts come first (Sections 3, 4): they do not use the classification (Section 5). Their proofs (Sections 3.3, 4) can be skipped: later sections use only the statements, apart from two proofs in Sections 6 and 7 that follow the steps of the proof of Theorem 4.1.

## 2 Equations, waves and what is counted

This section fixes the normalizations and records the form of the waves that are counted.

## 2.1 From the evolution equation to the profile equation

For a travelling wave  $u = w(\xi)$ ,  $\xi = x - ct$ , one integration turns (2.1) into (2.2):

$$u_t + uu_x + \sum_{j=1}^p b_j \partial_x^{j+1} u = 0, \quad b_p \neq 0, \quad (2.1)$$

$$\sum_{j=1}^p b_j w^{(j)} - cw + \frac{1}{2}w^2 + K = 0, \quad (2.2)$$

with a constant  $K$ . Let  $\kappa \neq 0$ , and let  $w_*$  be a constant solution of (2.2), a root of  $w^2/2 - cw + K$ . The substitution (2.3) turns (2.2) into the profile equation (1.1) with coefficients (2.4):

$$w(\xi) = w_* + b_p \kappa^p v(\kappa \xi), \quad (2.3)$$

$$a_j = \frac{b_j}{b_p} \kappa^{j-p} \quad (1 \leq j \leq p-1), \quad a_0 = \frac{w_* - c}{b_p \kappa^p}. \quad (2.4)$$

Thus  $a_1, \dots, a_{p-1}$  record the equation, up to the scale  $\kappa$ , and  $a_0$  records the speed of the wave relative to the background  $w_*$ . The scale is free at this stage. The period of the wave will fix it: if  $v$  is rational in  $e^z$ , then  $w$  is rational in  $e^{\kappa \xi}$ , so that  $\kappa$  is the rate of the wave.

The constant solutions of (1.1) are  $v = 0$  and  $v = -2a_0$ . They correspond to the two possible choices of the background  $w_*$ . Changing that choice replaces the pair  $(P, v)$  by

$$(P - 2P(0), v + 2P(0)), \quad (2.5)$$

which again satisfies (1.1), as a direct substitution shows. In the evolution equation the change of background is a Galilean shift  $u \mapsto u + k$ ,  $x \mapsto x - kt$ ; it changes the speed relative to the background, hence the sign of  $a_0$ . The two pairs describe the same solution  $u(x, t)$  of the same evolution equation. They coincide exactly when  $a_0 = 0$ . For example, the KdV soliton gives the pair  $P = \rho^2 - 1$ ,  $v = 3 \operatorname{sech}^2(z/2)$ . Measured from the other background it is  $P = \rho^2 + 1$  and  $v - 2$ , which does not tend to zero and is therefore not a rational-exponential pair.

## 2.2 Dispersive, dissipative and mixed equations

In (2.1) the term  $b_j \partial_x^{j+1} u$  is dispersive for even  $j$  and dissipative for odd  $j$ : for real coefficients it changes the phase, respectively the amplitude, of a Fourier mode (with the signs of (2.1), a positive coefficient of  $u_{xx}$  is gain). The equation is *purely dispersive* if  $b_j = 0$  for odd  $j$ . Then  $p$  is even,  $P$  is an even polynomial, and the equation is invariant under  $(x, t) \mapsto (-x, -t)$ . The equation is *purely dissipative* if  $b_j = 0$  for even  $j$ . Then  $p$  is odd,  $P - P(0)$  is an odd polynomial, and the equation is invariant under  $(x, u) \mapsto (-x, -u)$ ; the Nikolaevskiy equation ( $p = 5$ ) is an example, see Simbawa et al. [72]. All other equations are *mixed*.

The two pure subfamilies are described by one condition. We call a monic polynomial  $P$  of degree  $p$  *reflection-symmetric* if  $P - P(0)$  has the parity of  $\rho^p$ :

$$P(-\rho) = P(\rho) \quad (p \text{ even}), \quad P(\rho) + P(-\rho) = 2P(0) \quad (p \text{ odd}). \quad (2.6)$$

In the travelling-wave variable both symmetries are the reflection  $z \mapsto -z$ . If  $P$  is even and  $v(z)$  solves (1.1), so does  $v(-z)$ . If  $p$  is odd and  $P - P(0)$  is odd, so does  $-v(-z) - 2P(0)$ ; this is the reflection combined with the change of background (2.5).

### 2.3 Poles, and the form of one-pole waves

To count waves we need to know what they look like. Three facts suffice: a pole has order  $p$  and a fixed leading coefficient; with one pole per period, a wave rational in  $e^z$  is a polynomial of degree  $p$  in the logistic function  $Q$ , and an elliptic wave is a combination of  $\wp$  and its derivatives.

Throughout, a meromorphic function is single-valued and meromorphic on the whole complex plane. The order of a pole of a solution of (1.1) is forced by dominant balance. A pole of order  $r$  makes  $P(D)v$  of order  $r + p$  and  $v^2$  of order  $2r$ ; the two can cancel only if  $r = p$ . Comparing the leading coefficients gives, at a pole  $z_0$ ,

$$v(z) = \frac{c_*}{(z - z_0)^p} + O((z - z_0)^{1-p}), \quad c_* = (-1)^{p-1}(p-1)! s_p, \quad s_p = \frac{2(2p-1)!}{((p-1)!)^2}. \quad (2.7)$$

This holds for every  $P$ . The scale  $s_p$  of the normal forms of Sections 3 and 4 has the first values  $s_2 = 12$ ,  $s_3 = 60$ ,  $s_4 = 280$ ,  $s_6 = 5544$ , so that  $c_* = -12, 120, -1680$  for  $p = 2, 3, 4$ .

The meromorphic travelling waves in this paper are of three kinds: rational in  $z$ , rational in an exponential  $e^{\kappa z}$ , and elliptic; Theorem 5.1 says when there are no others. The period group of a wave is its *full* group of periods: trivial in the first case, generated by one period in the second, a lattice in the third. In the second case we choose  $\kappa$  so that  $2\pi i/\kappa$  generates this group. The wave has *one pole per period* if all its poles are translates of a single pole by the periods. A rational wave then has a single pole, and an elliptic wave one in each period parallelogram.

Let  $v = R(t)$ ,  $t = e^{\kappa z}$ , be a wave rational in an exponential, with one pole per period. The rational function  $R$  has finite limits at both  $t = 0$  and  $t = \infty$ . Indeed,  $P(D) = P(\kappa t \partial_t)$  multiplies each monomial  $t^m$  by a constant, whereas squaring doubles an extreme exponent: if  $R$  had a pole at either end,  $R^2$  would contain an extreme Laurent monomial that  $P(\kappa t \partial_t)R$  cannot cancel. Both limits are constant solutions of (1.1). After a translation, the only pole of  $R$  in  $0 < |t| < \infty$  is at  $t = -1$ , of order  $p$ . For  $\kappa = 1$  such a function is a polynomial of degree  $p$  in the logistic function  $Q = 1/(1 + e^{-z}) = (1 + \tanh(z/2))/2$ , with  $DQ = Q(1 - Q)$ , which has a simple pole with residue one at  $z = i\pi$  and the limits 0 and 1. By (2.7) the coefficient of  $Q^p$  is  $c_*$ ; the KdV soliton above is  $12Q(1 - Q)$ . Polynomial logistic representations and their use in constructing travelling waves are developed by Kudryashov [39, 56, 57], and applied to equations of the family up to  $p = 6$  by Ryabov et al. [71].

Such a wave is a *pulse* if its two limits coincide and a *front* otherwise. For a real rate and a real profile these are the solitary waves and the kinks of the evolution equation, and they are smooth and bounded on the real axis when the pole lies off it. In (2.3) we take  $w_*$  to be the limit as  $e^{\kappa z} \rightarrow 0$ . Then  $v \rightarrow 0$  at that end, and the other limit is 0 for a pulse and  $-2a_0$  for a front.

An elliptic wave whose only pole modulo its periods is at zero is a constant plus a linear combination of  $\wp, \wp', \dots, \wp^{(p-2)}$ , in which  $\wp^{(p-2)}$  has the coefficient  $-s_p$  (Section 4). Rational waves are not counted, but they govern what happens at infinity in both counts. A rational solution has no nonconstant polynomial part at infinity: such a part would have its degree doubled by  $v^2$ , while the linear differential expression cannot reach that degree. A rational wave whose only pole is at zero is therefore

$$v(z) = c_* z^{-p} + (\text{a polynomial of degree at most } p-1 \text{ in } z^{-1}). \quad (2.8)$$

The simplest one is  $v = c_* z^{-p}$ , for  $P = D^p$ ; it is given for every order by Kudryashov [45].

Conversely, a function of one of these forms that satisfies (1.1) is a nonconstant wave with one pole per period, because  $c_* \neq 0$ . Substitution turns (1.1) into finitely many polynomial equations in the coefficients of the form and of  $P$ , without any genericity assumption. They are instances of

the principal-part construction of Demina and Kudryashov [20, 21]; for a fixed operator it reduces to the coefficient comparison of Demina and Kudryashov [15], which we do not need.

## 2.4 What is counted

We count, over  $\mathbb{C}$ , pairs  $(P, v)$  in which  $P$  ranges over *all* monic polynomials of degree  $p$  and  $v$  solves (1.1). In a *rational-exponential pair*,  $v$  is rational in  $e^z$ , its poles are the points of  $i\pi + 2\pi i\mathbb{Z}$ , and  $v \rightarrow 0$  as  $e^z \rightarrow 0$ . In an *elliptic pair* on a lattice  $\Lambda$ ,  $v$  is elliptic with period lattice  $\Lambda$ , including its scale, and its poles are the points of  $\Lambda$ . Fixing the periods removes the scale  $\kappa$  in (2.3), and placing the pole removes translations. A wave and its mirror image are counted as two pairs when they are distinct (Section 6). Both pairs (2.5) of an elliptic wave are counted; for a rational-exponential wave the condition  $v \rightarrow 0$  selects one of them.

## 3 The count of rational-exponential pairs

We count the pairs  $(P, v)$  of Section 2 for all monic  $P$  of degree  $p$ . Whether an equation can have other meromorphic waves is a separate question (Section 5); nothing in this section or the next depends on the answer.

By Section 2.3, a normalized profile is a polynomial of degree  $p$  in  $Q = 1/(1 + e^{-z})$ , without constant term because  $v \rightarrow 0$  as  $e^z \rightarrow 0$ . Since  $D^j Q = (-1)^j j! Q^{j+1}$  plus lower powers of  $Q$ , the functions  $Q, DQ, \dots, D^{p-1}Q$  span the degree- $p$  polynomials in  $Q$  with zero constant term, and the leading pole coefficient (2.7) gives the unique form

$$v_A = s_p A(D)Q, \quad A(\rho) = \rho^d + \alpha_1 \rho^{d-1} + \dots + \alpha_d, \quad d = p - 1, \quad (3.1)$$

with the scale  $s_p$  of (2.7); the unknowns of the count are  $\alpha_1, \dots, \alpha_d$ . This is the standard singular-part operator form:  $Q = 1/2 + D \log \cosh(z/2)$ , so  $v_A$  is a constant plus a linear differential operator applied to  $\log \cosh(z/2)$ ; such forms, including the KS case, are discussed by Conte and Musette [6].

### 3.1 Discrete convolution and the matching equations

For polynomials  $F, G$ , let  $F \star G$  be the polynomial with  $(F \star G)(n) = \sum_{j=1}^{n-1} F(j)G(n-j)$  for all integers  $n \geq 1$ . It exists because, by the power-sum formulas, the right side is a polynomial in  $n$ . Write  $S_A = A \star A$ . For monic  $A$  of degree  $d$ ,

$$\deg S_A = 2d + 1, \quad [\rho^{2d+1}]S_A = \frac{(d!)^2}{(2d+1)!} = \frac{2}{s_p}; \quad (3.2)$$

the leading coefficient is the beta integral  $\int_0^1 t^d(1-t)^d dt$ .

**Proposition 3.1** (Polynomial divisibility). *The profile (3.1) solves (1.1) for some monic  $P$  if and only if  $A$  divides  $S_A$ . The operator is unique and is*

$$P_A = \frac{s_p S_A}{2A}. \quad (3.3)$$

*Proof.* In the half-plane  $\operatorname{Re} z < 0$ , the logistic function is a geometric series in  $e^z$ , and applying  $A(D)$  multiplies the  $n$ th term by  $A(n)$ :  $A(D)Q = \sum_{n \geq 1} (-1)^{n-1} A(n) e^{nz}$ . Squaring gives the rational-function identity

$$(A(D)Q)^2 = -S_A(D)Q. \quad (3.4)$$

The map  $F \mapsto F(D)Q$  is injective: its vanishing would give  $F(n) = 0$  for every positive integer  $n$ . Consequently (1.1) is equivalent to  $2P_A A = s_p S_A$ , which has a polynomial solution  $P_A$  exactly when  $A$  divides  $S_A$ . Equation (3.2) shows that  $P_A$  is monic of degree  $p$ .  $\square$

Divide  $S_A$  by the monic polynomial  $A$ :

$$S_A = AC_A + R_A, \quad R_A = \sum_{r=0}^{d-1} R_r(\alpha_1, \dots, \alpha_d) \rho^r. \quad (3.5)$$

The matching equations are  $R_0 = \dots = R_{d-1} = 0$ :  $d$  polynomial equations in the  $d$  unknowns  $\alpha_1, \dots, \alpha_d$ . Their solutions are in bijection with the normalized pairs  $(P_A, v_A)$ ; we write  $\mathcal{X}_p$  for the set of solutions. For  $p = 2$  and  $A = \rho + a$  there is the single equation  $R_0 = -a(a-1)(a+1)/6 = 0$  of the Introduction. Multiplicities in  $\mathcal{X}_p$  refer to these equations.

**Simple solutions and multiplicity.** A solution is *simple* if the Jacobian matrix  $(\partial R_r / \partial \alpha_j)$  is invertible there, like a simple root,  $f(a) = 0$ ,  $f'(a) \neq 0$ . Every isolated solution, that is, one that is not a limit of other solutions, has a *multiplicity*: the number of solutions near it of the system  $R_r = \varepsilon_r$  for generic small  $\varepsilon \in \mathbb{C}^d$ , as the double root of  $x^2 = 0$  becomes the two roots of  $x^2 = \varepsilon$ . It is one exactly when the solution is simple, and it equals the dimension over  $\mathbb{C}$  of the convergent power series at that point modulo the combinations  $\sum_r h_r R_r$  with power series  $h_r$ . For finitely many solutions, the sum of their multiplicities, the *count with multiplicity*, is the dimension of the polynomials modulo the polynomial combinations [2, Sections 4.3 and 5.2], [11, Chapter 4, §2]; it is the number of distinct pairs exactly when every solution is simple (see Section 8.2).

**How many solutions can there be?** Before counting, we ask how many solutions such a system can have at most. The answer is already  $N_p$ . Give  $\alpha_j$  weight  $j$  and  $\rho$  weight one, so that  $A$  is weighted-homogeneous of weight  $d$ : replacing the roots of  $A$  by  $\lambda$  times themselves multiplies  $\alpha_j$  by  $\lambda^j$ . The polynomial  $S_A$  has weight at most  $2d + 1$ , and its part of weight exactly  $2d + 1$  is the *continuous* convolution

$$H_A(\rho) = \int_0^\rho A(t)A(\rho - t) dt, \quad \int_0^\rho t^i(\rho - t)^j dt = \frac{i!j!}{(i+j+1)!} \rho^{i+j+1}. \quad (3.6)$$

Indeed these are the highest weighted terms of the power sums that define  $S_A$ . Monic division preserves the highest weighted part:  $A$  is weighted-homogeneous and monic, so each step of long division subtracts a multiple of  $A$  of the same weight as the term being removed. Consequently the matching equation  $R_r$  has weight at most  $2d + 1 - r$ , and

$$\text{the part of weight } 2d + 1 - r \text{ of } R_r \text{ is the coefficient of } \rho^r \text{ in } \text{rem}_A H_A. \quad (3.7)$$

In words, the discrete convolution  $S_A$  is a lower-order perturbation of the continuous convolution  $H_A$ , and far from the origin the matching equations look like the continuous problem  $A \mid H_A$ . Solutions of the matching equations could escape to infinity only along solutions of that problem. *Rigidity* ( $R_d$ ) is the statement that it has none except the trivial one:

$$A \text{ monic, } \deg A = d, \quad A \mid H_A \implies A(\rho) = \rho^d. \quad (3.8)$$

**Proposition 3.2** (Weighted Bézout bound). *Let  $p \geq 2$  and  $d = p - 1$ . The matching equations are  $d$  equations of weighted degrees at most  $2d + 1, 2d, \dots, d + 2$  in  $d$  unknowns of weights  $1, 2, \dots, d$ . Their isolated solutions, counted with multiplicity, number at most*

$$\frac{(d+2)(d+3) \cdots (2d+1)}{d!} = \binom{2d+1}{d} = \binom{2p-1}{p-1} = N_p.$$

*If rigidity ( $R_d$ ) holds, that is, if the parts of weight  $2d + 1 - r$  of the  $R_r$ , which by (3.7) are the coefficients of  $\text{rem}_A H_A$ , have no common zero other than the origin, then the matching equations have finitely many solutions, and counted with multiplicity there are exactly  $N_p$  of them. They are then  $N_p$  distinct solutions exactly when all of them are simple.*

For  $p = 2, 3$  the bound is  $3/1 = 3$  and  $(4 \cdot 5)/(1 \cdot 2) = 10$ . The substitution  $\alpha_j = y_j^j$  turns weights into ordinary degrees and counts every solution  $d!$  times; the proof, from Bézout's inequality for isolated solutions [66, Chapter VIII] and Bézout's theorem [11], is Theorem 2.1 of the companion note. The proof of Theorem 3.4 does not use the proposition: there the labels give the solutions, and rigidity is a by-product. The proposition is what remains at other orders (Section 9).

For waves, rigidity is a statement about rational solutions. The dictionary is the Laplace transform, restricted to polynomials.

**Lemma 3.3** (Laplace correspondence). *Let  $F \mapsto F^\sharp$  be the linear map on polynomials with  $(\rho^k)^\sharp = k! z^{-k-1}$ . Then  $D(F^\sharp) = (-\rho F)^\sharp$ , and  $F^\sharp G^\sharp = (F * G)^\sharp$  for the continuous convolution  $(F * G)(\rho) = \int_0^\rho F(t)G(\rho - t) dt$ ; in particular  $(A^\sharp)^2 = (H_A)^\sharp$ . Let  $P$  be monic of degree  $p$ , and let  $V$  be a nonconstant rational solution of (1.1) whose only pole is at  $z = 0$ . Then  $V$  vanishes at infinity,  $P(0) = 0$ , and*

$$V = (-1)^{p-1} s_p A^\sharp, \quad A \mid H_A, \quad P(\rho) = (-1)^p \frac{s_p}{2} \frac{H_A}{A}(-\rho),$$

*with  $A$  monic of degree  $p - 1$ . Conversely, every monic  $A$  of degree  $p - 1$  that divides  $H_A$  gives such a solution. The residue of  $V$  at zero is  $(-1)^{p-1} s_p A(0)$ .*

*Proof.* Both rules are checked on monomials; the second is the integral in (3.6). By (2.8),  $V = b + f$  with a constant  $b$  and with  $f \rightarrow 0$  at infinity. The constant part of (1.1) is  $b(P(0) + b/2) = 0$ , and the slowest decaying term of  $f$  enters the rest of the equation with the factor  $P(0) + b$ , which must vanish. Hence  $b = P(0) = 0$ . By (2.7) the pole has order  $p$  and leading coefficient  $c_* = (-1)^{p-1} (p-1)! s_p$ , so  $V = cA^\sharp$  with  $c = (-1)^{p-1} s_p$  and  $A$  monic of degree  $p - 1$ . By the two rules, (1.1) for  $V$  reads  $(cP(-\rho)A + c^2 H_A/2)^\sharp = 0$ , and the map  $\sharp$  is injective. This gives the divisibility, the formula for  $P$ , and the converse: the  $P$  defined by the formula is monic of degree  $p$ , because  $H_A$  has the leading coefficient (3.2) of  $S_A$ . The residue is the coefficient  $cA(0)$  of  $z^{-1}$ .  $\square$

For  $A = \rho^{p-1}$  the lemma gives  $V = c_* z^{-p}$  and  $P = D^p$ . Rigidity ( $R_{p-1}$ ) is therefore the statement that, for every monic  $P$  of degree  $p$ , equation (1.1) has no other nonconstant rational solution with a single pole.

## 3.2 The prime-index theorem

**Theorem 3.4** (Prime-index enumeration). *Let  $\ell = 2p - 1$  be prime, and put  $d = p - 1$ .*

- (a) *The matching equations have exactly  $\binom{\ell}{d} = N_p$  solutions over  $\mathbb{C}$ , and every solution is simple. Thus there are exactly  $N_p$  rational-exponential pairs, and they are distinct.*

- (b) *The coefficients of every solution are algebraic numbers, and they are  $\ell$ -integral: they have no  $\ell$  in a denominator and can be reduced modulo  $\ell$ , in the way in which  $1/11$  becomes 1 and  $i$  becomes 2 or 3 modulo 5. Once a consistent way of reducing algebraic numbers is fixed (Section 3.3), reduction modulo  $\ell$  is a bijection between the solutions and the  $d$ -element subsets  $L$  of the field  $\mathbb{F}_\ell$  of integers modulo  $\ell$ :*

$$\overline{A}(\rho) = \prod_{s \in L} (\rho - s). \quad (3.9)$$

- (c) *Rigidity ( $R_d$ ) holds.*

We call  $L$  the *label* of the pair. It depends on the way of reducing algebraic numbers, an embedding fixed in the proof, only as  $i$  is 2 or 3 modulo 5: another embedding gives a solution the label that the first one gives to a conjugate solution. Whether a label contains 0, and the labels of solutions with rational coefficients, do not depend on it. The labels describe reductions of algebraic coefficients; they do not assert that the complex roots of  $A$ , or the characteristic roots of  $P_A$ , are integers.

**Example: order three.** Let  $p = 3$ ,  $s_3 = 60$ , and  $A(\rho) = \rho^2 + a\rho + b$ , so that  $(a, b) = (\alpha_1, \alpha_2)$ . The profile and monic operator are

$$\frac{v_A}{60} = 2Q^3 - (a + 3)Q^2 + (a + b + 1)Q, \quad (3.10)$$

$$P_A(\rho) = \rho^3 + 4a\rho^2 + (a^2 + 19b)\rho - a^3 + 7ab - 5a - 30b. \quad (3.11)$$

The remainder of  $S_A$  is  $(\widehat{R}_1\rho + \widehat{R}_0)/30$  with

$$\widehat{R}_0 = 30R_0 = ab(a^2 - 7b + 5), \quad \widehat{R}_1 = 30R_1 = a^4 - 8a^2b + 11b^2 + 10b - 1. \quad (3.12)$$

We solve  $\widehat{R}_0 = \widehat{R}_1 = 0$  by cases on the factors of  $\widehat{R}_0$ . If  $b = 0$ , then  $a^4 = 1$ . If  $a = 0$ , then  $(b + 1)(11b - 1) = 0$ . In the remaining case  $a^2 = 7b - 5$ , and the second equation becomes  $4(b - 2)(b - 3) = 0$ . This exhausts the system and gives Table 2. The determinant  $\det \partial(\widehat{R}_0, \widehat{R}_1)/\partial(a, b)$  is nonzero at every point, so all  $N_3 = 10$  pairs are simple, as Theorem 3.4(a) states for  $2p - 1 = 5$ .

Table 2: All ten rational-exponential pairs at order three. The label is the set of roots of  $A$  modulo 5; for the two complex pairs it depends on the choice of  $i$  modulo 5. For the equation of the Introduction,  $(b_1, b_2, b_3) = (1, \sigma, 1)$ , the coefficients  $B = 4a$  of  $\rho^2$  and  $\mu = a^2 + 19b$  of  $\rho$  in (3.11) are  $\sigma/\kappa$  and  $1/\kappa^2$  by (2.4), so  $\sigma^2 = B^2/\mu = 16a^2/(a^2 + 19b)$ . In the rows  $(\pm i, 0)$  and  $(0, -1)$ ,  $\mu < 0$  and the rate  $\kappa$  is imaginary. The last column describes  $v_A$  on the real axis, at unit rate.

| $a$     | $b$  | Labels               | $ \sigma $     | Determinant | Real profiles |
|---------|------|----------------------|----------------|-------------|---------------|
| $\pm 1$ | 0    | $\{0, 4\}, \{0, 1\}$ | 4              | -24         | Pulses        |
| $\pm i$ | 0    | $\{0, 2\}, \{0, 3\}$ | 4              | -16         | Not real      |
| 0       | -1   | $\{1, 4\}$           | 0              | 144         | Front         |
| 0       | 1/11 | $\{2, 3\}$           | 0              | 576/121     | Front         |
| $\pm 3$ | 2    | $\{3, 4\}, \{1, 2\}$ | $12/\sqrt{47}$ | -144        | Fronts        |
| $\pm 4$ | 3    | $\{2, 4\}, \{1, 3\}$ | $16/\sqrt{73}$ | 384         | Fronts        |

The ten labels are the ten two-element subsets of  $\{0, 1, 2, 3, 4\}$ . By (3.10),  $v_A/60 \rightarrow b$  as  $z \rightarrow +\infty$ , so a pair is a pulse exactly when  $b = 0$ . The map  $(a, b) \mapsto (-a, b)$  is the mirror image of

the Introduction, and it replaces a label  $L$  by  $-L$  (Section 6). Eight of the ten pairs have real coefficients, and all eight are smooth and bounded on the real axis: two pulses and six fronts. For example,  $A = (\rho - 1)(\rho - 2)$  gives  $v_A = 120Q^3$  and  $P_A = (\rho - 3)(\rho - 4)(\rho - 5)$ . These are the classical solutions in hyperbolic functions [32–35, 62], tabulated completely by Kudryashov and Zargaryan [43, Table 1] and Conte and Musette [7, Table 1]. Their tables contain six cases because they identify the two signs of the dispersion coefficient: the four mirror pairs  $(\pm a, b)$  and the two pairs with  $a = 0$ . Exhaustion of this list follows from Eremenko [23]; Demina [17] classifies all irreducible algebraic invariants of the same profile equation, and later rederivations by ansatz methods coincide with these waves [42]. The pairs  $(\pm 1, 0)$  and  $(\pm i, 0)$  belong to the same equations, those with  $\sigma^2 = 16$ , at the rates  $\kappa = \pm 1$  and  $\kappa = \pm i$  (Section 7). For a real equation the second wave is thus periodic in  $x$ , and its real form has poles on the real axis.

### 3.3 Proof of the prime-index theorem

The idea was described in the Introduction. A reader who accepts the theorem can go to Section 3.4: the rest of the paper uses only its statement, and the proof of Proposition 3.6 uses (3.14) and fact (V) below once more.

**Reduction modulo  $\ell$  of algebraic numbers.** We use four standard facts. Rational numbers without  $\ell$  in the denominator form a ring  $\mathbb{Z}_{(\ell)}$ . Think of  $\ell$  as a small parameter  $\varepsilon$ : the valuation is the order of a quantity in  $\varepsilon$ , and reduction modulo  $\ell$  sets  $\varepsilon = 0$ .

(V) *Valuation and reduction.* Let  $K$  be an algebraic closure of the field  $\mathbb{Q}_\ell$  of  $\ell$ -adic numbers. It is algebraically closed, contains  $\mathbb{Q}$ , and carries the  $\ell$ -adic valuation  $\text{ord}: K \rightarrow \mathbb{Q} \cup \{\infty\}$  [31, Chapter III], [67, Chapter II, Theorem (4.8)]: the exponent of  $\ell$  in a nonzero rational number, infinite only at 0, with

$$\text{ord}(xy) = \text{ord}(x) + \text{ord}(y), \quad \text{ord}(x + y) \geq \min\{\text{ord}(x), \text{ord}(y)\}. \quad (3.13)$$

We fix an embedding of the field  $\overline{\mathbb{Q}} \subset \mathbb{C}$  of algebraic numbers in  $K$ , unique up to an automorphism of  $\overline{\mathbb{Q}}$ , and regard algebraic numbers as elements of  $K$ . Those  $x \in K$  with  $\text{ord}(x) \geq 0$  are  $\ell$ -integral. They form a ring  $\mathcal{O} \supset \mathbb{Z}_{(\ell)}$ , and *reduction modulo  $\ell$* ,  $x \mapsto \bar{x}$ , maps  $\mathcal{O}$  onto its residue field  $\mathbb{k} \supset \mathbb{F}_\ell$ : it respects sums and products, it is the usual reduction on  $\mathbb{Z}_{(\ell)}$ , and  $\bar{x} = 0$  exactly when  $\text{ord}(x) > 0$ . Polynomials are reduced coefficient by coefficient. Division by a monic polynomial uses only sums and products of coefficients: it commutes with substituting values for the  $\alpha_j$  and with reduction, and it stays within  $\mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$  or  $\mathcal{O}$ . A monic polynomial with  $\ell$ -integral roots has  $\ell$ -integral coefficients.

(H) *Hensel's lemma.* Let  $G = (G_1, \dots, G_d)$  be polynomials in  $d$  unknowns with coefficients in  $\mathbb{Z}_{(\ell)}$ . If  $\bar{a} \in \mathbb{F}_\ell^d$  solves the reduced system and its Jacobian determinant is not zero there, then exactly one  $a \in \mathcal{O}^d$  has  $G(a) = 0$  and reduction  $\bar{a}$ . This is Newton's method from an integer vector that reduces to  $\bar{a}$ , just as a simple root at  $\varepsilon = 0$  continues uniquely to a power series in  $\varepsilon$ : the multivariable Hensel lemma of Conrad [5], applied in the finite extensions of  $\mathbb{Q}_\ell$  in  $K$ , which are complete [67, Chapter II, Theorem (4.8)].

(N) *The weak Nullstellensatz.* Polynomial equations with rational coefficients that have a solution in  $\mathbb{C}^d$  have one in  $\overline{\mathbb{Q}}^d$  [12, Chapter 4, §1].

(T) *Finite systems.* If  $d$  polynomial equations in  $d$  unknowns with rational coefficients have only finitely many solutions in  $K^d$ , then these lie in  $\overline{\mathbb{Q}}^d$ , and they are also the solutions in  $\mathbb{C}^d$ , with the Jacobian determinant nonzero at the same ones. Indeed, by finiteness a nonzero polynomial

in each  $x_i$  alone is a combination of the equations [12, Chapter 5, §3]; finding one of bounded degree is a linear problem over  $\mathbb{Q}$ , so one with rational coefficients exists, and it confines  $x_i$  to its roots, in  $K$  as in  $\mathbb{C}$ .

*Proof of Theorem 3.4.* The proof has four steps: a congruence for  $S_A$  modulo  $\ell$ ; the solution of the reduced equations; integrality, a dominant balance showing that every solution can be reduced, and then reduces to one of them; and lifting, Newton's method in the form of Hensel's lemma, showing that each of them comes from exactly one solution. Solutions are first taken in  $K^d$ , with  $K$  as in (V); the last step returns to  $\mathbb{C}$ .

*Step 1: the congruence.* Induction using  $(D - n)Q^n = -nQ^{n+1}$  gives  $(n - 1)!Q^n = (-1)^{n-1} \prod_{j=1}^{n-1} (D - j)Q$ . Write  $(A(D)Q)^2 = \sum_{n=2}^{2p} \gamma_n Q^n$ . Each  $\gamma_n$  is an integer polynomial in the coefficients of  $A$ , and  $\gamma_{2p} = (d!)^2$ . By (3.4),

$$S_A(\rho) = \sum_{n=2}^{2p} \frac{(-1)^n \gamma_n}{(n-1)!} \prod_{j=1}^{n-1} (\rho - j).$$

Only the last denominator,  $(2p - 1)! = \ell!$ , contains  $\ell$ . Consequently  $\ell S_A$  has coefficients in  $\mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$ , and

$$\ell S_A(\rho) \equiv \frac{(d!)^2}{(\ell - 1)!} \prod_{j=1}^{\ell} (\rho - j) \equiv (-1)^d (\rho^\ell - \rho) \pmod{\ell}. \quad (3.14)$$

The scalar follows by pairing  $j$  with  $\ell - j$  in  $(\ell - 1)!$ . The product is  $\rho^\ell - \rho$  because, by Fermat's little theorem, every residue  $s$  modulo  $\ell$  satisfies  $s^\ell = s$ . The right side does not depend on  $A$ : this is what makes the prime index special. Taking the parts of weight  $\ell = 2d + 1$  on both sides gives the corresponding congruence for the continuous convolution,

$$\ell H_A(\rho) \equiv (-1)^d \rho^\ell \pmod{\ell}. \quad (3.15)$$

Both congruences are between polynomials in  $\rho$  and the  $\alpha_j$  with coefficients in  $\mathbb{Z}_{(\ell)}$ , so they may be evaluated at any  $\ell$ -integral values of the  $\alpha_j$  and of  $\rho$ , and then reduced.

*Step 2: the reduced equations.* By (V) the scaled remainders  $\ell R_r$  have coefficients in  $\mathbb{Z}_{(\ell)}$ , and by (3.14) their reductions are the coefficients of the remainder of  $(-1)^d (\rho^\ell - \rho)$  on division by  $\bar{A}$ . The reduced equations therefore say that  $\bar{A}$  divides  $\rho^\ell - \rho$ . Now  $\rho^\ell - \rho = \prod_{s \in \mathbb{F}_\ell} (\rho - s)$ , and polynomials over a field factor uniquely. Over any field that contains  $\mathbb{F}_\ell$ , in particular over  $\mathbb{k}$ , the monic divisors of degree  $d$  are therefore the  $\binom{\ell}{d}$  polynomials (3.9), and their coefficients lie in  $\mathbb{F}_\ell$ . Each of them is a simple solution of the reduced equations. Write  $\rho^\ell - \rho = \bar{A} \bar{C}$ . For a variation  $h$  with  $\deg h < d$ , the derivative of the remainder is, up to the nonzero scalar  $(-1)^d$ ,

$$h \longmapsto -\text{rem}_{\bar{A}}(\bar{C}h), \quad (3.16)$$

because the dividend in (3.14) does not vary with  $A$ . Since  $\rho^\ell - \rho$  has no repeated factor,  $\bar{A}$  and  $\bar{C}$  have no common factor. If  $\bar{A}$  divides  $\bar{C}h$ , it therefore divides  $h$ , and then  $h = 0$  because  $\deg h < d$ . The linear map (3.16) is thus injective, hence invertible.

*Step 3: integrality and rigidity.* A solution could have  $\ell$  in a denominator, and then it could not be reduced. We show that this does not happen. The argument is a dominant balance, with  $\ell$  in the role of the small parameter: a solution that is large is rescaled to size one, and the leading-order problem, which is the continuous one, then leads to a contradiction. For  $\lambda \neq 0$  and

a monic polynomial  $A$  of degree  $d$ , put  $A_\lambda(\rho) = \lambda^{-d}A(\lambda\rho)$ . This is the monic polynomial whose roots are those of  $A$  divided by  $\lambda$ , and its coefficients are  $\alpha'_j = \alpha_j/\lambda^j$ . A polynomial  $T$  in the  $\alpha_j$  and  $\rho$  that is weighted-homogeneous of weight  $w$  satisfies  $T(\alpha; \lambda\rho) = \lambda^w T(\alpha'; \rho)$ .

Let  $A$  be a solution of the matching equations with coefficients in  $K$ , and suppose that some root of  $A$  is not  $\ell$ -integral. Let  $\lambda$  be a root of least valuation, so that  $\theta = \text{ord}(\lambda) < 0$ , and put  $B = A_\lambda$ . The roots of  $B$  are  $\ell$ -integral, hence so are its coefficients  $\alpha'_j$ , by (V). Decompose  $\ell S_A = \sum_{w \leq \ell} T^{(w)}$  into its parts of weight  $w$ . They have coefficients in  $\mathbb{Z}_{(\ell)}$  by Step 1, and  $T^{(\ell)} = \ell H_A$ . Since  $A$  divides  $S_A$  and  $A(\lambda) = 0$ , we have  $S_A(\lambda) = 0$ . Dividing this equation by  $\lambda^\ell$  gives

$$\ell H_B(1) = - \sum_{w < \ell} \lambda^{w-\ell} T^{(w)}(\alpha'; 1).$$

Each term on the right has positive valuation, because  $\text{ord}(\lambda^{w-\ell}) = (w-\ell)\theta > 0$  and  $T^{(w)}(\alpha'; 1)$  is  $\ell$ -integral. By (3.13) the right side reduces to zero. The left side reduces to  $(-1)^d \neq 0$ , by (3.15) evaluated at  $\alpha = \alpha'$  and  $\rho = 1$ . This contradiction shows that every root of  $A$  is  $\ell$ -integral, and so is every coefficient. The solution can be reduced, and by Step 2 its reduction is one of the polynomials (3.9).

Rigidity is proved in the same way. Suppose that  $A \neq \rho^d$  is monic of degree  $d$ , with complex coefficients, and divides  $H_A$ . Since  $H_{A_\lambda}(\rho) = \lambda^{-\ell} H_A(\lambda\rho)$ , every  $A_\lambda$  has the same property, so we may assume that one of the coefficients  $\alpha_j$  equals 1. By (N), applied to the equations  $\text{rem}_A H_A = 0$  together with this normalization, we may assume that the coefficients of  $A$  lie in  $\overline{\mathbb{Q}} \subset K$ . Let  $\lambda$  be a root of  $A$  of least valuation; it is not zero, because  $A \neq \rho^d$ . As before,  $B = A_\lambda$  has  $\ell$ -integral coefficients. Since  $A$  divides  $H_A$  and  $A(\lambda) = 0$ , we have  $0 = \ell H_A(\lambda) = \lambda^\ell \ell H_B(1)$ . But  $\ell H_B(1)$  reduces to  $(-1)^d \neq 0$ , by (3.15). This contradiction proves (c).

*Step 4: lifting.* The system  $(\ell R_r)_{0 \leq r < d}$  has coefficients in  $\mathbb{Z}_{(\ell)}$ , and by Step 2 each of the  $\binom{\ell}{d}$  reduced solutions lies in  $\mathbb{F}_\ell^d$  and is simple. By Hensel's lemma (H), each of them is the reduction of exactly one solution in  $\mathcal{O}^d$ . By Step 3 every solution in  $K^d$  lies in  $\mathcal{O}^d$ , and its reduction is one of the reduced solutions. Hence reduction modulo  $\ell$  is a bijection between the solutions in  $K^d$  and the reduced solutions. There are exactly  $\binom{\ell}{d}$  solutions in  $K^d$ , and the Jacobian determinant is nonzero at each of them, because its reduction is nonzero. By (T), the matching equations have the same  $\binom{\ell}{d}$  solutions over  $\mathbb{C}$ , with coordinates in  $\overline{\mathbb{Q}}$ , all simple. This proves (a) and (b); part (c) was proved in Step 3.  $\square$

*Remark 3.5* (Formal verification). Parts (a) and (c) of Theorem 3.4 have been verified in Lean 4 against the Mathlib library, both for the matching equations and for the pairs  $(P, v)$  themselves (Appendix A). The formal proof follows the route above, and like the proof above it normalizes the roots of  $A$ .

### 3.4 Reading the labels: pulses and fronts

As  $z \rightarrow +\infty$  we have  $Q \rightarrow 1$  and  $D^j Q \rightarrow 0$  for  $j \geq 1$ , so  $F(D)Q \rightarrow F(0)$  for every polynomial  $F$ . The limits of  $v_A$  as  $z \rightarrow \mp\infty$  are therefore 0 and  $s_p A(0)$ : pairs with  $A(0) = 0$  are pulses, and the others are fronts. The pulses are needed separately, because the elliptic pairs of Section 4 degenerate onto them.

Letting  $z \rightarrow +\infty$  in (3.4) gives  $S_A(0) = -A(0)^2$ . Since  $S_A(0) = A(0)C_A(0) + R_0$  by (3.5), this gives the polynomial identity

$$R_0 = -A(0)(A(0) + C_A(0)). \quad (3.17)$$

On the hyperplane  $\alpha_d = A(0) = 0$  the equation  $R_0 = 0$  thus holds identically, and the pulses are the solutions of the *pulse equations*

$$R_1 = \cdots = R_{d-1} = 0 \quad \text{in the unknowns } \alpha_1, \dots, \alpha_{d-1}, \quad \alpha_d = 0. \quad (3.18)$$

We write  $\mathcal{Z}_p$  for the set of their solutions; multiplicities of points of  $\mathcal{Z}_p$  refer to the pulse equations.

**Proposition 3.6** (Pulses and fronts). *Let  $\ell = 2p - 1$  be prime. A pair is a pulse if and only if its label contains 0. Consequently exactly  $S_p := \binom{2p-2}{p-2}$  of the  $N_p$  pairs are pulses, and the remaining  $\binom{2p-2}{p-1}$  are fronts. Every pulse is a simple solution of the pulse equations (3.18), and  $P_A(0) \neq 0$  at every pulse (these two facts are used in Section 4). These are counts over  $\mathbb{C}$ .*

In Table 2 the pulses are the four pairs with  $b = 0$ , and their labels are the four that contain 0.

*Proof.* Let  $A$  be a solution with label  $L$ . If  $A(0) = 0$ , then  $\bar{A}(0) = 0$ , so  $0 \in L$ . Conversely, let  $0 \in L$ . At the solution  $A$  we have  $\ell S_A = A \cdot \ell C_A$ . By (V) and (3.14),  $\ell C_A$  is  $\ell$ -integral and reduces to  $(-1)^d \bar{C}$ , where  $\bar{C} = (\rho^\ell - \rho)/\bar{A}$ . Since  $\rho$  divides  $\bar{A}$  and  $\rho^\ell - \rho$  has no repeated factor,  $\bar{C}(0) \neq 0$ . The number  $\ell A(0)$  reduces to zero. Hence  $\ell(A(0) + C_A(0))$  reduces to  $(-1)^d \bar{C}(0) \neq 0$ . So  $A(0) + C_A(0) \neq 0$ , and  $R_0 = 0$  gives  $A(0) = 0$ , by (3.17): the pair is a pulse. The number of  $d$ -element subsets of  $\mathbb{F}_\ell$  that contain 0 is  $\binom{\ell-1}{d-1} = S_p$ .

At a pulse, (3.17) shows that the row of the Jacobian matrix of the matching equations that belongs to  $R_0$  has a single entry that can be different from zero, namely  $-C_A(0)$  in the column of  $\alpha_d$ . The Jacobian matrix is invertible, by Theorem 3.4(a). Hence  $C_A(0) \neq 0$ , so that  $P_A(0) = s_p C_A(0)/2 \neq 0$ , and the minor obtained by deleting that row and that column is invertible. This minor is the Jacobian matrix of the pulse equations. (For  $d = 1$  there are no pulse equations, and only  $C_A(0) \neq 0$  is asserted.)  $\square$

## 4 The count of elliptic pairs

We count the elliptic pairs of Section 2 on a fixed lattice  $\Lambda \subset \mathbb{C}$ , with  $g_i = g_i(\Lambda)$ . Neither reflection nor a lattice automorphism is factored out (so that  $M_3 = 8$  below, and not fewer). Two lattices differ by a rescaling exactly when they have the same invariant  $j(\Lambda) = 1728g_2^3/(g_2^3 - 27g_3^2)$ .

We write elliptic functions through the Weierstrass relations

$$Y^2 = 4X^3 - g_2X - g_3, \quad \mathcal{D}X = Y, \quad \mathcal{D}Y = 6X^2 - \frac{g_2}{2}. \quad (4.1)$$

On a lattice, that is, for  $g_2^3 \neq 27g_3^2$ , they are realized by  $X = \wp(z; g_2, g_3)$ ,  $Y = \wp'(z; g_2, g_3)$  and  $\mathcal{D} = d/dz$  [70, Chapter 23]. We use  $X$  and  $Y$  because the formulas must keep their meaning for the degenerate parameters  $g_2^3 = 27g_3^2$ , which belong to no lattice. By (4.1), every polynomial in  $X$  and  $Y$  has a unique normal form  $F(X) + YG(X)$ . We say that  $X^k Y^\epsilon$ ,  $\epsilon \in \{0, 1\}$ , has pole order  $2k + 3\epsilon$ ; on a lattice this is its pole order at zero.

An elliptic function with one pole per period has zero residue there, and  $\wp, \wp', \dots, \wp^{(p-2)}$  have poles of orders  $2, 3, \dots, p$  at zero. Subtracting a suitable combination of them from a one-pole solution leaves an entire elliptic function, hence a constant. Since  $D^{p-2}\wp = (-1)^p(p-1)!z^{-p} + O(1)$ , the leading coefficient (2.7) gives every profile the classical form

$$V = -s_p \mathcal{D}^{p-2}X + \sum_{j=0}^{p-3} u_j \mathcal{D}^j X + b. \quad (4.2)$$

Its only pole modulo  $\Lambda$  is at zero, so  $\Lambda$  is its full period lattice. The matching equations  $\mathcal{E}_p(g_2, g_3)$ , in the unknowns  $a_0, \dots, a_{p-1}, u_0, \dots, u_{p-3}$  and  $b$ , are obtained by formal substitution of (4.2) in (1.1), using only (4.1): the left side is reduced to its normal form, and the coefficients of  $F$  and  $G$  are set to zero. They are polynomials in the unknowns and in  $g_2, g_3$ , defined for every  $(g_2, g_3) \in \mathbb{C}^2$ , the degenerate parameters included; on a lattice their solutions are the elliptic pairs. We write  $\mathcal{E}_p(g_2, g_3)$  also for the set of solutions; multiplicity and simplicity are as in Section 3.

**Theorem 4.1** (Fixed-lattice elliptic enumeration). *Let  $2p - 1$  be prime.*

- (a) *On every lattice the elliptic pairs are finite in number, and counted with multiplicity there are*

$$M_p := 2 \binom{2p-2}{p-2} = 2S_p \quad (4.3)$$

*of them.*

- (b) *There is a proper algebraic subset  $\mathcal{N}$  of the  $(g_2, g_3)$ -plane outside which all of them are simple. The lattices in  $\mathcal{N}$  are those whose invariant  $j(\Lambda)$  lies in a finite set, which depends on  $p$ . Thus a generic lattice, that is, a lattice whose invariant avoids these values, admits exactly  $M_p$  distinct elliptic pairs.*

**The smallest case.** For  $p = 2$  the profile is  $V = -12X + b$ , and the left side of (1.1) reduces to  $-12a_1Y - 12(a_0 + b)X + 6g_2 + a_0b + \frac{1}{2}b^2$ . The complete elliptic system is therefore

$$a_1 = 0, \quad v = -12\wp - a_0, \quad a_0^2 = 12g_2. \quad (4.4)$$

No one-pole elliptic wave has a nonzero Burgers coefficient: both pairs belong to the KdV equation, in agreement with  $M_2 = 2$ , and they are the two backgrounds (2.5) of one cnoidal wave. They are distinct and simple for  $g_2 \neq 0$ ; for  $g_2 = 0, g_3 \neq 0$ , they coincide, and the lattice carries one pair of multiplicity two. Here  $\mathcal{N}$  is the line  $g_2 = 0$ , and the finite set of (b) is  $\{0\}$ ; at order three it is  $\{0, -300\}$  (Section 8.2).

**No escape to infinity.** When the parameters of a polynomial system with as many equations as unknowns vary, its solutions may merge or separate. As long as they remain finite in number, their number, counted with multiplicity, can change only if a solution moves off to infinity. Such a solution, rescaled to size one, converges to a nonzero solution of the leading parts of the equations, as in Step 3 of the proof of Theorem 3.4 with the ordinary absolute value in place of the  $\ell$ -adic one. Here the leading parts are the equations with both periods infinite ( $\wp \rightarrow z^{-2}$ ; the cusp  $g_2 = g_3 = 0$ ), and by rigidity they have only the trivial solution. So nothing escapes, and the number of pairs, counted with multiplicity, is the same for all  $(g_2, g_3)$ . We find it with one period infinite ( $\wp$  becomes a hyperbolic function; the node): there every elliptic pair becomes a pulse on one of its two backgrounds, and every pulse on either background arises exactly once, so  $M_p = 2S_p$ . The pulses are simple, and simplicity spreads to a generic lattice.

**Lemma 4.2** (Conservation of number). *Let  $E_1, \dots, E_n$  be polynomials in unknowns  $x = (x_1, \dots, x_n)$  and parameters  $t = (t_1, \dots, t_m)$ , weighted-homogeneous for some positive integer weights of the  $x_j$  and the  $t_k$ . Suppose that the leading system  $E_i(x; 0) = 0, 1 \leq i \leq n$ , has no solution in  $\mathbb{C}^n$  other than  $x = 0$ . Then:*

- (i) *for every  $t$  the system  $E(x; t) = 0$  has finitely many solutions, and they stay bounded while  $t$  stays bounded;*

- (ii) the number of solutions, counted with multiplicity, is the same for every  $t \in \mathbb{C}^m$ ;
- (iii) the  $t$  for which some solution is not simple form an algebraic subset of  $\mathbb{C}^m$ , that is, the common zero set of finitely many polynomials in  $t$ .

**Sketch of proof.** (i) Modulo the equations, every polynomial is a combination of finitely many monomials in  $x$  with coefficients polynomial in  $t$  (induction on the weight, as the  $t_k$  have positive weight). Hence each  $x_j$  satisfies a monic polynomial equation with coefficients polynomial in  $t$ , which gives finiteness and boundedness. (ii) By Rouché's theorem in several variables, the number of zeros in a ball, counted with multiplicity, is a degree on the bounding sphere and does not change under a deformation without zeros on the sphere [65, Appendix B], [2, Sections 5.2 and 5.4]. (iii) By elimination theory [12, Chapter 3, §1], the monic polynomials of (i) make the projection an algebraic set. The details are in the companion note, together with a second, algebraic proof of (i)–(ii) by flatness.

Three operations preserve, up to isomorphism, the space of power series modulo the equations, and hence multiplicities (Section 3): an invertible affine change of unknowns; invertible linear combinations of the equations with constant coefficients; the elimination of an unknown that one equation gives as a polynomial in the others.

*Proof of Theorem 4.1. Step 1: eliminating the operator.* The coefficient at pole order  $2p$  vanishes by the choice of  $s_p$ . The coefficients at orders  $2p-1, \dots, p$  determine  $a_{p-1}, \dots, a_0$  successively: at each step the coefficient of the new  $a_j$  is the nonzero rational leading coefficient of  $D^j V$ . Each  $a_j$  is thus a polynomial in the remaining unknowns and in  $g_2, g_3$ , and the elimination does not change multiplicities. The residual has pole order at most  $p-1$  and zero residue, since no normal form has pole order one, so it lies in the span of 1 and the  $\mathcal{D}^j X$ ,  $0 \leq j \leq p-3$ . Its coefficients in this basis are  $p-1$  equations  $E_1 = \dots = E_{p-1} = 0$  in the  $p-1$  profile unknowns  $u_j, b$ .

*Step 2: weights.* Give  $D, X, g_2, g_3$  the weights 1, 2, 4, 6 and the profile the weight  $p$ ; this is the usual homogeneity of  $\wp$  under a rescaling of  $z$ . Then  $u_j$  has weight  $p-j-2$ ,  $b$  has weight  $p$ , and the  $E_i$  are weighted-homogeneous polynomials in the  $u_j, b, g_2, g_3$ , of weights

$$\begin{array}{c|c} \text{variables} & 1, 2, \dots, p-2, p \\ \text{equations} & p+1, p+2, \dots, 2p-2, 2p. \end{array} \quad (4.5)$$

The Bézout number of this table is  $(p+1) \cdots (2p-2)(2p)/((p-2)!p) = 2 \binom{2p-2}{p-2} = M_p$  (at  $p=3$ :  $4 \cdot 6/(1 \cdot 3) = 8$ ), in agreement with the count below (companion note). Since  $g_2, g_3$  have positive weight, setting  $g_2 = g_3 = 0$  gives the parts of highest weight of the  $E_i$  in the profile unknowns.

*Step 3: the cusp.* At the cusp  $g_2 = g_3 = 0$  the relations (4.1) are satisfied by  $X = z^{-2}$ ,  $Y = -2z^{-3}$  and  $\mathcal{D} = d/dz$ , so a solution of the cusp system gives a nonconstant rational solution  $V$  of (1.1) whose only pole is at zero. By Lemma 3.3,  $V$  vanishes at infinity, so that  $b = 0$ ; moreover  $a_0 = P(0) = 0$ , and  $V = (-1)^{p-1} s_p A^\sharp$  for a monic  $A$  of degree  $p-1$  that divides  $H_A$ . By rigidity, which holds by Theorem 3.4(c),  $A = \rho^{p-1}$ . Thus  $V = c_* z^{-p}$ , and all the unknowns in (4.2) vanish, because the functions  $D^j z^{-2}$  have different pole orders: the leading equations have no nonzero common zero.

*Step 4: conservation of number.* By Steps 1–3, Lemma 4.2 applies to  $E_1, \dots, E_{p-1}$ , with the profile unknowns as  $x$  and  $t = (g_2, g_3)$ ; the leading system is the cusp system. By (i), every system  $\mathcal{E}_p(g_2, g_3)$  has finitely many solutions, which is the finiteness in (a). By (ii), their number, counted with multiplicity, is the same at every point of the  $(g_2, g_3)$ -plane, the degenerate parameters

included. By (iii), the parameters at which some solution is not simple form an algebraic subset  $\mathcal{N}$  of the plane.

*Step 5: the node.* To find the common number, and to show that  $\mathcal{N}$  is not the whole plane, consider the nodal parameters, for which the relations (4.1) are satisfied by polynomials in  $Q$ :

$$g_2 = \frac{1}{12}, \quad g_3 = -\frac{1}{216}; \quad (4.6)$$

$$X = Q^2 - Q + \frac{1}{12}, \quad Y = (2Q - 1)Q(1 - Q), \quad \mathcal{D} = D \quad \text{with} \quad DQ = Q(1 - Q).$$

This is the degenerate  $\wp$ -function with the single period  $2\pi i$ , with its pole moved to  $z = i\pi$ , in the logistic coordinate. A monomial  $X^k Y^\epsilon$  becomes a polynomial in  $Q$  of degree  $2k + 3\epsilon$ , which is also its pole order. These degrees are all different, so the monomials remain linearly independent, and the nodal matching equations are exactly the conditions for (1.1) to hold as an identity between polynomials in  $Q$ . Since  $X = \frac{1}{12} - DQ$  and  $\mathcal{D}^j X = -D^{j+1}Q$  for  $j \geq 1$ , the profile (4.2) becomes  $V = \beta + v_A$ , where  $A(\rho) = -s_p^{-1} \sum_{j=0}^{p-2} u_j \rho^{j+1}$  with  $u_{p-2} = -s_p$ , and  $\beta = b + u_0/12$ . Here  $A$  is monic of degree  $d = p - 1$  with  $A(0) = 0$ , and  $\beta$  is the common limit of  $V$  at the two ends. With the coefficients  $\alpha_j$  of  $A$  as in (3.1), the passage from  $(u_0, \dots, u_{p-3}, b)$  to  $(\alpha_1, \dots, \alpha_{d-1}, \beta)$  is an invertible affine change of unknowns with rational coefficients.

In these unknowns the constant term of  $P(D)V + V^2/2$  is  $P(0)\beta + \beta^2/2$ , and the remaining terms are  $(P + \beta)(D)v_A + v_A^2/2$ . By the proof of Proposition 3.1 the latter equals  $(s_p/2)[2(P + \beta)A - s_p S_A](D)Q$ , and the elimination of Step 1 is the division of  $S_A$  by  $A$ : it gives  $P + \beta = s_p C_A/2$ , a polynomial in the unknowns, which we still denote by  $P_A$ , and it leaves  $-(s_p^2/2)R_A(D)Q$ . Since  $A(0) = 0$ , the coefficient  $R_0$  vanishes identically, by (3.17). With  $P(0) = P_A(0) - \beta$ , the nodal matching equations are therefore invertible linear combinations, with constant coefficients, of

$$R_1 = \dots = R_{d-1} = 0, \quad \beta(\beta - 2P_A(0)) = 0, \quad (4.7)$$

together with  $P = P_A - \beta$ : the pulse equations (3.18) and one quadratic equation for the background. The two systems have the same solutions with the same multiplicities.

By Proposition 3.6 the pulse equations have exactly  $S_p$  solutions, all simple, with  $P_A(0) \neq 0$  at each of them. For each of them the quadratic has the two simple roots  $\beta = 0$  and  $\beta = 2P_A(0)$ , and the Jacobian matrix of (4.7) is block triangular. For the nodal parameters the system thus has exactly  $2S_p$  solutions, all simple. By Step 4 the number of solutions counted with multiplicity is  $2S_p = M_p$  for all  $(g_2, g_3)$ , which proves (a), and the nodal parameters do not belong to  $\mathcal{N}$ . So  $\mathcal{N}$  is a proper algebraic subset of the plane, and outside  $\mathcal{N}$  every solution is simple.

The  $E_i$  and their Jacobian determinant are weighted-homogeneous, so  $\mathcal{N}$  is invariant under  $(g_2, g_3) \mapsto (\lambda^4 g_2, \lambda^6 g_3)$ ,  $\lambda \neq 0$ ; on lattices this is the rescaling  $\Lambda \mapsto \lambda^{-1} \Lambda$ . Hence, if a polynomial vanishes on  $\mathcal{N}$ , so does each of its weighted-homogeneous components, and the proper algebraic subset  $\mathcal{N}$  lies in the zero set of a nonzero polynomial that is a monomial in  $g_2, g_3$  times a product  $\prod_i (g_2^3 - c_i g_3^2)$ . On the lattices this zero set consists of finitely many values of  $j$ :  $j = 0$  on  $g_2 = 0$ ,  $j = 1728$  on  $g_3 = 0$ , and  $j = 1728c_i/(c_i - 27)$  on  $g_2^3 = c_i g_3^2$ . Two lattices with the same  $j$  differ by a rescaling, so  $\mathcal{N}$  contains both or neither: the lattices in  $\mathcal{N}$  are exactly those whose invariant lies in a finite set. This proves (b).  $\square$

**Two backgrounds.** The factor two in  $M_p$  counts the two backgrounds (2.5); changing the background changes  $P$  and the speed, not the wave or the lattice. Outside  $\mathcal{N}$  the two backgrounds of every wave are distinct, because a solution with  $a_0 = 0$  is never simple: Step 1 gives  $a_0 + b$  as

a polynomial in the  $u_j$  and  $g_2, g_3$ , and the residual equations contain  $b$  only through  $a_0^2$ , in the constant one; so their Jacobian determinant has the factor  $a_0$ . On a generic lattice the  $M_p$  pairs are therefore  $M_p/2$  elliptic waves, up to translations and Galilean shifts, each on both backgrounds; on special lattices the two may coalesce, as in (4.4) for  $g_2 = 0$  and in Section 8.2.

The elliptic pairs of purely dispersive equations,  $2F_p$  of the  $M_p$ , are counted in Corollary 6.4, once the symmetry of their profiles is established; at order two they are both pairs of (4.4), and  $M_2 = 2F_2 = 2$ .

## 5 Are these all the meromorphic waves? The classification theorem

The counts did not ask whether an equation can have other meromorphic travelling waves. For two classes of equations it cannot. The proof is a local analysis of Painlevé type, combined with Eremenko's theorem. A by-product, Corollary 5.3, identifies the two pure subfamilies (Section 6).

**Theorem 5.1** (Classification). *Assume either that  $p$  is odd or that  $P$  is an even polynomial; that is, the evolution equation has even order or is purely dispersive. Every nonconstant meromorphic solution of (1.1) is of one of the three kinds of Section 2.3: rational in  $z$ , rational in  $e^{\kappa z}$  for some  $\kappa \neq 0$ , or elliptic. Every pole has order  $p$ , and the solution has one pole per period: its poles are the translates of one pole by the full group of periods.*

*For fixed  $P$  of odd degree, there is at most one nonconstant solution with its pole at zero. For even  $P$ , the same statement holds after fixing the value of the first integral (5.5) below, the energy. The only entire solutions are  $v = 0$  and  $v = -2a_0$ .*

*Remark 5.2* (The hypothesis cannot be dropped: the Fisher family). The travelling waves of nonzero speed of the Fisher equation satisfy (1.1) with  $p = 2$  and a mixed operator, the profile equation of KdV–Burgers. At the special speed of Ablowitz and Zeppetella [1] it has a one-parameter family of solutions, given in Kudryashov [38], Conte et al. [8, Lemma 4.4] and McCue et al. [64], and for the KdV–Burgers equation itself in Kudryashov [47]; see also Kudryashov [36]. In our normalization it is

$$v(z) = -12\kappa^2 e^{2\kappa z} \wp(e^{\kappa z} - 1; 0, g_3), \quad P(D) = D^2 - 5\kappa D + 6\kappa^2, \quad \kappa \neq 0,$$

as  $(D^2 - 5\kappa D + 6\kappa^2)[t^2 U(t)] = \kappa^2 t^4 U''(t)$ ,  $t = e^{\kappa z}$ , shows. For  $g_3 = 0$  it degenerates to the rational-exponential Ablowitz–Zeppetella front, for  $\kappa = 1$  the front in the first row of (8.2). For generic  $g_3 \neq 0$  it is globally meromorphic and of none of the three kinds; further such examples are discussed by Ng and Wu [68, Remark 1.11]. For mixed operators of even degree  $p \geq 4$ , the classification theorem above does not decide whether solutions of other kinds occur.

**Earlier work.** Theorem 5.1 is a uniform version of results that exist in pieces; for the analytic background see Eremenko [22]. Odd  $p$  is contained in Yuan et al. [75, Theorem 1], with related results in Huang et al. [26], Yuan et al. [76] and a fifth-order instance in Dang [14]. For even  $P$ , fixing a first integral to remove the freedom at the resonance is the device of Eremenko et al. [24, Lemmas 3–4] for  $P = D^p + a_0$ . It is stated as a general step by Conte et al. [9, Section 3.2] and Conte et al. [10, Section 5], and applied to the Kawahara equation [19, 74], the generalized Bretherton equation [20, 61] and the seventh-order KdV equation [13]; the familiar Kawahara first integral is the case  $m = 2$  of (5.5). Theorem 5.1 extends the device to every even  $P$ , intermediate derivatives included, with the explicit coefficient (5.7). Equations with two dominant monomials,

or without the finiteness property (finitely many Laurent expansions at a pole), are treated by other methods in Demina [16, 18]; see also Belova [3, Sections 5.2–5.3]. These include the classification of algebraic first-order invariants by Puiseux expansions, not only reconstruction from a prescribed Laurent principal part. For the mixed quadratic fourth-order equation, Belova [3, Section 5.6, Theorem 5.8] gives all elliptic and rational-exponential solutions, including the one-pole conclusion, despite a free Laurent coefficient at resonance 12. That result assumes membership in those function classes; it does not assert that every global meromorphic solution belongs to them. The quadratic equation there contains a constant forcing term, which can be removed by a shift over  $\mathbb{C}$ , changing  $a_0$  as in our normalization.

### 5.1 The Laurent series at a pole

At a pole, placed at zero, write

$$v(z) = \sum_{n \geq 0} c_n z^{n-p}, \quad c_0 = c_*, \quad c_n = 0 \quad (n < 0); \quad (5.1)$$

the pole order and  $c_0$  are those of (2.7), for every  $P$ . Comparing the coefficients of  $z^{n-2p}$  in (1.1) gives, for  $n \geq 1$ ,

$$I_p(n) c_n = - \sum_{j=0}^{p-1} a_j \prod_{i=1}^j (n - 2p + i) c_{n-p+j} - \frac{1}{2} \sum_{k=1}^{n-1} c_k c_{n-k}, \quad (5.2)$$

$$I_p(n) = \prod_{j=p}^{2p-1} (n - j) + c_*. \quad (5.3)$$

The product in  $I_p$  comes from  $D^p$  acting on  $z^{n-p}$ , and  $c_*$  from the cross term of  $c_n z^{n-p}$  with the leading pole in  $v^2/2$ . The right side of (5.2) contains only  $c_k$  with  $k < n$ , so it determines  $c_n$  whenever  $I_p(n) \neq 0$ . The positive integer zeros of  $I_p$  are the *resonances* (Fuchs indices): there  $c_n$  is left undetermined if the right side vanishes, and no expansion exists otherwise. By (2.7),  $c_*$  has the sign  $(-1)^{p-1}$ , and  $|c_*| = 2p(p+1) \cdots (2p-1)$  is twice the absolute value of the product at  $n = 0$ .

Let  $p$  be odd, so that  $c_* > 0$ . The product in (5.3) is negative with absolute value less than  $c_*/2$  for  $1 \leq n < p$ , zero for  $p \leq n \leq 2p-1$ , and positive for  $n \geq 2p$ . Hence  $I_p(n) \neq 0$  for every integer  $n \geq 1$ : there is no resonance, and exactly one formal Laurent expansion. Theorem 2 of Eremenko [23] states that an autonomous polynomial differential equation with finitely many formal pole expansions and a single monomial of highest total degree has all its meromorphic solutions among the three kinds. In (1.1) the only monomial of degree two is  $v^2/2$ , so for odd  $p$  both hypotheses hold, for all  $a_j$ .

### 5.2 The energy argument for a purely dispersive equation

Let  $p = 2m$  and  $P(D) = D^{2m} + \sum_{j=0}^{m-1} a_{2j} D^{2j}$ . Now  $c_* < 0$ . The product in (5.3) is at most  $|c_*|/2$  for  $0 \leq n < p$ , zero for  $p \leq n \leq 2p-1$ , and strictly increasing for  $n \geq 2p$ , with the value  $(p+1)(p+2) \cdots (2p) = -c_*$  at  $n = 3p$ . So the only resonance is  $n = 3p$ , and only  $c_{3p}$  can be free. If it is free, there is a one-parameter family of formal expansions, and Eremenko's theorem does not apply to (1.1) itself. The remedy is a first integral whose value determines  $c_{3p}$ .

For  $j \geq 1$  put

$$\mathcal{B}_j(v) = \sum_{i=0}^{j-2} (-1)^i v^{(i+1)} v^{(2j-1-i)} + \frac{(-1)^{j-1}}{2} (v^{(j)})^2, \quad (5.4)$$

where the sum is empty when  $j = 1$ . Termwise differentiation gives  $D\mathcal{B}_j = v'v^{(2j)}$ . Since  $P$  contains only even derivatives, the *energy*

$$\mathcal{H}_P(v) = \mathcal{B}_m(v) + \sum_{j=1}^{m-1} a_{2j} \mathcal{B}_j(v) + \frac{a_0}{2} v^2 + \frac{1}{6} v^3 \quad (5.5)$$

satisfies

$$D\mathcal{H}_P(v) = v' \left( P(D)v + \frac{1}{2} v^2 \right), \quad (5.6)$$

and is constant on each solution. It is the energy in the usual sense: an even  $P(D)$  is formally self-adjoint, so (1.1) is an Euler–Lagrange equation, and  $\mathcal{H}_P$  is the quantity conserved by its invariance under translation in  $z$ .

The resonance is always compatible: if the series (5.1) through  $c_{3p-1}$  leaves the residual  $\Omega z^p + \dots$  in (1.1), then  $v'$  times it has residue  $-pc_*\Omega$ ; by (5.6) it is a derivative, which has no residue. So  $\Omega = 0$ , and  $c_{3p}$  is free.

But the energy determines  $c_{3p}$ : on the formal solution

$$\mathcal{H}_P = -pc_* I'_p(3p) c_{3p} + (\text{a constant that depends only on } P). \quad (5.7)$$

The terms of (5.5) with  $a_{2j}$  contain fewer derivatives than  $\mathcal{B}_m$ , so in them  $c_{3p}$  multiplies only positive powers of  $z$ ; it enters the constant term only through  $\mathcal{B}_m(v) + v^3/6$ , in the cross terms with the leading pole. Linearize (5.6) for  $P = D^p$  about its solution  $c_* z^{-p}$ , with perturbation  $z^{n-p}$ : if the energy term is  $\omega(n)z^{n-3p}$ , then  $(n-3p)\omega(n) = -pc_* I'_p(n)$ , and  $n \rightarrow 3p$  gives the coefficient in (5.7). It is not zero: the product in (5.3) equals  $-c_*$  at  $n = 3p$ , so  $I'_p(3p) = -c_* \sum_{j=p+1}^{2p} 1/j > 0$ . Thus the energy fixes  $c_{3p}$ , and (5.2) gives the later coefficients.

Eremenko's theorem now applies, not to (1.1) but to the level equation, with  $P$  and the value of  $\mathcal{H}_P(v)$  fixed. It is autonomous and polynomial, with the single monomial  $v^3/6$  of highest degree, and has exactly one formal pole expansion: differentiating it and dividing by the nonzero series  $v'$  returns (1.1), so every expansion belongs to the family above, and (5.7) selects  $c_{3p}$ . Every meromorphic solution of (1.1) lies on one energy level and so is of one of the three kinds.

### 5.3 The period group and the entire solutions

In either case of the theorem, let  $z_1, z_2$  be two poles of one solution. The functions  $v(z + z_1)$  and  $v(z + z_2)$  satisfy the same equation and, in the even case, have the same energy. Their Laurent series at zero coincide, so  $v(z + z_1) = v(z + z_2)$  by the identity theorem: every difference of poles is a period. Conversely, every period translates a pole to a pole, so the solution has one pole per period. The period group is discrete, since  $v$  is not constant. The same argument proves the uniqueness statements of the theorem: two nonconstant solutions with the same operator, the same energy in the even case, and a pole at zero have the same Laurent series there.

An entire solution of one of the three kinds is an entire elliptic function, hence constant, or a polynomial in  $z$ , or a Laurent polynomial in  $e^{\kappa z}$ . The last two are constant as well, because squaring doubles an extreme nonzero exponent (Section 2.3). The constant solutions are  $v = 0$  and  $v = -2a_0$ . This proves the theorem.

The same uniqueness shows that, in the two pure subfamilies, every wave inherits the reflection symmetry of its equation.

**Corollary 5.3** (Waves are symmetric about their poles). *Let  $v$  be a nonconstant meromorphic solution of (1.1) and let  $z_0$  be one of its poles.*

- (i) *If  $P$  is even, then  $v(z_0 + \zeta) = v(z_0 - \zeta)$ .*
- (ii) *If  $p$  is odd and  $P - P(0)$  is odd, then  $v(z_0 + \zeta) + P(0) = -(v(z_0 - \zeta) + P(0))$ .*

*Consequently a solution that is rational in an exponential is a pulse in case (i). In case (ii) its two limits add up to  $-2P(0)$ , so it is a front unless  $P(0) = 0$ .*

The KdV soliton  $3 \operatorname{sech}^2(z/2)$  is even about its pole  $i\pi$ . For the Kuramoto–Tsuzuki front, the row  $(0, 1/11)$  of Table 2,  $v + P(0) = \frac{15}{11}(11\tau^3 - 9\tau)$  with  $\tau = \tanh(z/2)$  is odd about  $i\pi$ .

*Proof.* In case (i) the function  $v(z_0 - \zeta)$  solves the same equation, has a pole at  $\zeta = 0$ , and has the same energy as  $v$ , because every term of (5.4) and (5.5) contains an even total number of derivatives. Fixed-energy uniqueness gives (i). In case (ii) the function  $-v(z_0 - \zeta) - 2P(0)$  solves the same equation, by Section 2, and has a pole at  $\zeta = 0$ . For odd  $p$  the Laurent expansion at a pole is unique, which gives (ii). The two limits exist by Section 2.3, and the statements about them follow by letting  $\zeta$  tend to the two ends.  $\square$

## 6 Mirror images and the two pure subfamilies

This section counts the pairs of the two pure subfamilies. They are the pairs that coincide with their mirror image  $\iota A$ , where for monic  $A$  of degree  $d = p - 1$

$$(\iota A)(\rho) = (-1)^d A(-\rho), \quad \text{that is,} \quad \alpha_j \mapsto (-1)^j \alpha_j. \quad (6.1)$$

**Proposition 6.1** (Reflection). *For every monic  $A$  of degree  $d$ ,*

$$S_{\iota A}(\rho) = -S_A(-\rho) - 2A(0)A(-\rho), \quad R_{\iota A}(\rho) = -R_A(-\rho). \quad (6.2)$$

*Hence  $\iota$  maps solutions of the matching equations to solutions, and simple solutions to simple ones. At a point of  $\mathcal{X}_p$ ,*

$$v_{\iota A}(z) = (-1)^p (v_A(-z) - s_p A(0)), \quad P_{\iota A}(\rho) = (-1)^p (P_A(-\rho) + s_p A(0)). \quad (6.3)$$

By (6.3),  $\iota A$  is indeed the mirror image of  $A$ : the wave  $v_A$  read in the opposite direction, with the zero limit moved back to the end  $e^z \rightarrow 0$  and, for odd  $p$ , the sign changed. For  $p = 2$ ,  $\iota$  exchanges the fronts  $12Q(2 - Q)$  and  $-12Q^2$  of (8.2):  $12(1 - Q)(1 + Q) - 12 = -12Q^2$ .

*Proof.* Since  $Q(-z) = 1 - Q(z)$  and  $D$  changes sign under  $z \mapsto -z$ , we have  $(A(D)Q)(-z) = A(0) - A(-D)Q = A(0) - (-1)^d (\iota A)(D)Q$ , which is the first formula in (6.3). Evaluating (3.4) at  $-z$ , and using it for  $\iota A$  on the left, gives

$$A(0)^2 - 2A(0)A(-D)Q - S_{\iota A}(D)Q = -S_A(0) + S_A(-D)Q.$$

The constant terms cancel because  $S_A(0) = -A(0)^2$  (Section 3.4), and the injectivity of  $F \mapsto F(D)Q$  gives the first identity in (6.2). Substituting  $S_A = AC_A + R_A$  at  $-\rho$ , with  $A(-\rho) =$

$(-1)^d(\iota A)(\rho)$ , turns it into a division of  $S_{\iota A}$  by the monic polynomial  $\iota A$ , which is unique. So  $R_{\iota A}(\rho) = -R_A(-\rho)$  and  $C_{\iota A}(\rho) = (-1)^p(C_A(-\rho) + 2A(0))$ . The first relation reads  $R_r(\iota\alpha) = (-1)^{r+1}R_r(\alpha)$ : the invertible linear map (6.1) takes each matching equation to plus or minus a matching equation, hence the assertion about solutions. The second relation and (3.3) give  $P_{\iota A}$ .  $\square$

A pair is fixed by  $\iota$  when  $\alpha_j = 0$  for every odd  $j$ ; we write  $\mathcal{X}_p^{\text{sym}}$  for the set of these pairs. On this subspace  $\iota A = A$ , so  $R_A$  is an odd polynomial by (6.2), and the equations  $R_r = 0$  with  $r$  even hold identically. The *matching equations of the subfamily* are the equations  $R_r = 0$ ,  $r$  odd, in the unknowns  $\alpha_j$ ,  $j$  even; their solutions are the pairs of  $\mathcal{X}_p^{\text{sym}}$ . Put

$$F_p = \left( \begin{matrix} p-1 \\ \lfloor (p-1)/2 \rfloor \end{matrix} \right) = 1, 2, 3, 6, 10, 20 \quad (p = 2, \dots, 7). \quad (6.4)$$

**Theorem 6.2** (Count in the two pure subfamilies). (i) *For every  $p$ , a pair  $A$  coincides with its mirror image,  $\iota A = A$ , if and only if  $P_A$  is reflection-symmetric. Thus these pairs are exactly the rational-exponential pairs of the purely dispersive equations when  $p$  is even, and of the purely dissipative equations when  $p$  is odd.*

(ii) *Let  $\ell = 2p - 1$  be prime. If  $A$  has label  $L$ , then  $\iota A$  has label  $-L$ . Hence  $\iota A = A$  if and only if  $L = -L$ , and there are exactly  $F_p$  such pairs. Each of them is a simple solution of the matching equations of its subfamily.*

(iii) *The rational-exponential pairs of a purely dispersive equation are pulses, for every even  $p$ . Those of a purely dissipative equation are fronts: when  $2p - 1$  is prime, and for pairs with real coefficients at every odd  $p$ .*

In Table 2 the two pairs with  $a = 0$  are the  $F_3 = 2$  pairs of the Kuramoto–Sivashinsky equation without dispersion; their labels  $\{1, 4\}$  and  $\{2, 3\}$  satisfy  $L = -L$ . Both are fronts; the pulses  $(\pm 1, 0)$  need dispersion.

*Remark 6.3* (Every order). At every order the isolated solutions of the matching equations of the subfamily, counted with multiplicity, number at most  $F_p$ , the Bézout number of this system (companion note, Theorem 4.1).

*Proof of Theorem 6.2.* (i). If  $\iota A = A$ , then (6.3) gives  $P_A(\rho) = (-1)^p(P_A(-\rho) + s_p A(0))$ . For even  $p$ ,  $A$  is odd, so  $A(0) = 0$  and  $P_A$  is even. For odd  $p$ ,  $P_A(\rho) + P_A(-\rho) = -s_p A(0)$ :  $P_A$  is reflection-symmetric, and

$$s_p A(0) = -2P_A(0) \quad (p \text{ odd}). \quad (6.5)$$

Conversely, let  $P = P_A$  be reflection-symmetric. By Corollary 5.3 at the pole  $i\pi$ , and by the period  $2\pi i$ , we have  $v_A(-z) = v_A(z)$  if  $p$  is even and  $-v_A(-z) = v_A(z) + 2P(0)$  if  $p$  is odd. So (6.3) becomes  $v_{\iota A} = v_A + c$  with a constant  $c$ , and  $c = 0$  because both profiles tend to zero as  $z \rightarrow -\infty$ . Hence  $\iota A = A$  by the uniqueness of the form (3.1).

(ii). The map (6.1) negates the roots of  $\bar{A}$ , so the label of  $\iota A$  is  $-L$ . A label determines its pair (Theorem 3.4), so  $\iota A = A$  exactly when  $L = -L$ . The nonzero residues modulo  $\ell = 2d + 1$  form  $d$  pairs  $\{s, -s\}$ , and a set  $L = -L$  of  $d$  elements consists of  $\lfloor d/2 \rfloor$  of these pairs, together with 0 if  $d$  is odd. So there are  $\binom{d}{\lfloor d/2 \rfloor} = F_p$  such sets.

Differentiating  $R_r(\iota\alpha) = (-1)^{r+1}R_r(\alpha)$  at a fixed point shows that  $\partial R_r / \partial \alpha_j = 0$  unless  $j \equiv r + 1$  modulo two. So the Jacobian matrix of the full system splits into two square blocks. It is invertible, so both blocks are, and one of them is the Jacobian of the equations of the subfamily.

(iii). For even  $p$ ,  $A(0) = 0$  as in (i). Let  $p$  be odd. If  $2p - 1$  is prime, a label with  $L = -L$  has an even number  $d$  of elements, so it does not contain 0, and the pair is a front by Proposition 3.6. For general odd  $p$ , let a point of  $\mathcal{X}_p^{\text{sym}}$  with real coefficients be a pulse,  $A(0) = 0$ . Then  $P_A(0) = 0$  by (6.5). Now  $v_A$  is real on the real axis and tends to zero with all its derivatives at both ends, so integrating (1.1) along the real axis, with  $P_A(0) = 0$ , gives  $\int v_A^2 dz = 0$ , which is impossible.  $\square$

**Corollary 6.4** (Elliptic pairs of purely dispersive equations). *Let  $p$  be even, with  $2p - 1$  prime. On every lattice, the elliptic pairs whose operator is an even polynomial are those with  $u_j = 0$  for odd  $j$  in (4.2), and counted with multiplicity there are*

$$2F_p \tag{6.6}$$

of them. On a generic lattice they are distinct.

*Proof.* By Corollary 5.3 an even operator has an even profile, and (4.2) is even exactly when  $u_j = 0$  for odd  $j$ . Conversely, for an even profile the odd part of (1.1) is  $\sum_{j \text{ odd}} a_j D^j V = 0$ ; the  $D^j V$  have distinct pole orders, so  $a_j = 0$  for odd  $j$ . We follow the proof of Theorem 4.1 on the subspace  $u_j = 0$ ,  $j$  odd. There the residual of Step 1 is even, so only the unknowns and the equations of even weight in (4.5) remain,  $p/2$  of each. The omitted leading forms vanish identically on the subspace, so by Step 3 the others have only the trivial common zero, and Lemma 4.2 applies to the restricted system; multiplicities refer to it. For the nodal parameters (4.6) reflection is  $Q \mapsto 1 - Q$ , and an even profile is  $\beta + v_A$  with  $\iota A = A$ . The computation of Step 5 on this subspace gives (4.7) with  $A \in \mathcal{X}_p^{\text{sym}}$ : the matching equations of the subfamily, and the quadratic for  $\beta$ . By Theorem 6.2 there are  $F_p$  such  $A$ , all pulses, each a simple solution of these equations, and  $P_A(0) \neq 0$  by Proposition 3.6. So the restricted nodal system has  $2F_p$  solutions, all simple, and the rest of Step 5 applies without change.  $\square$

For odd  $p$  there is no such count. The profile of a purely dissipative equation is  $-P(0)$  plus a function odd about the pole (Corollary 5.3), and its matching system has one equation more than unknowns. For  $p = 3, 5, 7$  an exact computation (`elliptic_dissipative.py`, Appendix A) finds no solution on a sample lattice, hence there is none on a generic lattice (the entries 0 of Table 1): rigidity holds at these orders (Section 9), so by Lemma 4.2 and the elimination in the proof of its part (iii) the parameters with such pairs form an algebraic subset of the  $(g_2, g_3)$ -plane, invariant under rescaling as in the proof of Theorem 4.1(b). At  $p = 3$  such pairs occur exactly when  $g_2 = 0$ : a purely dissipative operator has  $B = 0$  in (8.3), and then  $P = D^3 + a_0$  and  $g_2 = 0$ . At  $p = 5$  elliptic waves of the Nikolaevskiy equation on special lattices are given by Kudryashov [45] and Kudryashov and Demina [49].

## 7 What the count means for a single equation

We read the count for one equation (2.1) with fixed complex coefficients. It is invariant under translations and Galilean shifts.

**Proposition 7.1** (The waves of a fixed equation). *Let  $\kappa \neq 0$ . Up to a translation and a Galilean shift, the travelling waves of (2.1) that are rational in  $e^{\kappa(x-ct)}$ , with one pole modulo  $(2\pi i/\kappa)\mathbb{Z}$ , are*

$$u = b_p \kappa^p v_A(\kappa(x - ct)), \quad c = -b_p \kappa^p P_A(0), \tag{7.1}$$

where  $A$  ranges over the points of  $\mathcal{X}_p$  that satisfy

$$\frac{b_j}{b_p} = \kappa^{p-j} [\rho^j] P_A, \quad 1 \leq j \leq p-1. \quad (7.2)$$

The data  $(A, \kappa)$  and  $(\iota A, -\kappa)$  describe the same wave, read in the two directions (Section 6).

*Proof.* Such a wave has a finite limit as  $e^{\kappa(x-ct)} \rightarrow 0$  (Section 2.3), and a Galilean shift makes it zero. Then  $w_* = 0$  in (2.3), and (2.4) becomes (7.2) and the formula for  $c$ . After a translation  $v$  has the normalization of Section 2, so  $v = v_A$  and  $P = P_A$  by Proposition 3.1. The last assertion follows from (6.3): the two waves differ by the constant  $b_p \kappa^p s_p A(0)$ , which is a Galilean shift.  $\square$

Given a pair, (7.2) are  $p-1$  conditions on the  $p-1$  ratios  $b_j/b_p$  and the one unknown  $\kappa$ : a pair imposes  $p-2$  conditions on the equation and fixes the rate. A pair with real coefficients gives a real wave of a real equation, smooth on the real axis, exactly when  $\kappa$  is real. Let  $\mathbf{b} = (b_{p-1}/b_p, \dots, b_1/b_p)$  and  $\mathbf{a}(A) = ([\rho^{p-1}]P_A, \dots, [\rho^1]P_A)$  be the coefficient vectors of the equation and of  $P_A$ . By (7.2),  $\mathbf{b}$  is  $\mathbf{a}(A)$  with its  $k$ th entry multiplied by  $\kappa^k$ . As  $\kappa$  varies these vectors trace a curve  $\mathbb{C}^* \mathbf{a}(A)$ , and rescaling  $x$  moves an equation along it. Several pairs may give the same curve.

**Corollary 7.2** (Solvable equations). *Assume that the matching equations have finitely many solutions. This holds when  $2p-1$  is prime, and whenever rigidity holds (Proposition 3.2), hence for every  $p \leq 573$  (Section 9).*

- (i) Equation (2.1) has a travelling wave that is rational in an exponential, with one pole per period, if and only if  $\mathbf{b}$  lies on one of the curves  $\mathbb{C}^* \mathbf{a}(A)$ ,  $A \in \mathcal{X}_p$ . There are at most  $(N_p + F_p)/2$  such curves.
- (ii) If  $\mathbf{b} \neq 0$ , there are finitely many such waves up to translations and Galilean shifts.
- (iii) If  $p \geq 3$ , the curves form a set of dimension at most one in  $\mathbb{C}^{p-1}$ , and a generic equation has no such wave.

*Proof.* Part (i) restates Proposition 7.1. For the bound,  $\mathbf{a}(\iota A) = (-1) \cdot \mathbf{a}(A)$  by (6.3), so  $A$  and  $\iota A$  give the same curve. The number of classes  $\{A, \iota A\}$  is half the number of points plus half the number of fixed points. The solutions are finite in number, hence isolated, so these two numbers are at most  $N_p$  and  $F_p$  (Proposition 3.2, Remark 6.3). For (ii), if  $\mathbf{b} \neq 0$  and  $\mathbf{b} \in \mathbb{C}^* \mathbf{a}(A)$ , some  $[\rho^j]P_A$  is nonzero, and (7.2) leaves at most  $p-j$  values of  $\kappa$ . Part (iii) is a dimension count.  $\square$

When  $\mathbf{b} = 0$  the equation is scale-invariant, and each of its waves occurs at every rate. At order two there are two curves, and both carry waves:  $\mathbf{b} = 0$  is KdV, with its solitons, and each equation with  $\mathbf{b} \neq 0$  has one front, its rate fixed by the Burgers coefficient (Section 8.1).

At  $p = 3$ , write  $P = D^3 + BD^2 + \mu D + a_0$ , so that  $\mathbf{b} = (B, \mu)$ . Rescaling multiplies  $B$  by  $\kappa$  and  $\mu$  by  $\kappa^2$ , so the curves are  $B^2 = \sigma^2 \mu$  with the origin removed, where  $\sigma$  is the dispersion coefficient of the Introduction. By the caption of Table 2, the ten pairs give

$$\sigma^2 = \frac{16a^2}{a^2 + 19b} \in \{0, 16, 144/47, 256/73\}, \quad |\sigma| \in \{0, 4, 12/\sqrt{47}, 16/\sqrt{73}\}, \quad (7.3)$$

the four values of Kudryashov [33, Eq. (39)]. The bound  $(N_3 + F_3)/2 = 6$  is not attained: the two pairs with  $a = 0$  share  $\sigma = 0$ , and  $(\pm 1, 0)$  and  $(\pm i, 0)$  share  $\sigma^2 = 16$ . For coefficients off these curves, Kudryashov and Kochanov [58] construct approximate solutions close to the exact ones.

**Corollary 7.3** (Generic equations with  $p$  odd). *Let  $p \geq 5$  be odd and assume rigidity ( $R_{p-1}$ ), which holds when  $2p - 1$  is prime and for every  $p \leq 573$ . The coefficient vectors  $\mathbf{b} \in \mathbb{C}^{p-1}$  for which (2.1) has a nonconstant meromorphic travelling wave lie in an algebraic subset of dimension at most two. In particular a generic equation has none.*

*Proof.* By Theorem 5.1, such a wave is rational, rational in an exponential, or elliptic, with one pole per period. Take  $\kappa = 1$  in (2.3), so that  $[\rho^j]P = b_j/b_p$  for  $1 \leq j \leq p-1$ . The rational-exponential waves give a set of dimension at most one (Corollary 7.2). A rational wave is, after a translation, of the form (2.8); by Lemma 3.3 it comes from a monic  $A$  of degree  $p-1$  that divides  $H_A$ , and rigidity gives  $A = \rho^{p-1}$ , hence  $P = D^p$  and  $\mathbf{b} = 0$ . For the elliptic waves, Steps 1–3 of the proof of Theorem 4.1 use only rigidity, so by Lemma 4.2(i) every system  $\mathcal{E}_p(g_2, g_3)$  has finitely many solutions. The solutions for all  $(g_2, g_3)$  together thus form an algebraic set of dimension at most two, and so does the closure of its image under  $P \mapsto \mathbf{b}$ .  $\square$

*Remark 7.4* (Order three). At  $p = 3$  the argument gives the exact answer: these  $\mathbf{b} = (B, \mu)$  form the union of the four curves  $B^2 = \sigma^2 \mu$  of (7.3). The elliptic waves require  $B^2 = 16\mu$  (Section 8.2), which is one of the four, and the point  $\mathbf{b} = 0$  of the rational wave lies on all of them.

The same dimension count applies to the one-pole waves of every order  $p \geq 3$  for which rigidity holds: a generic equation has no travelling wave that is rational, rational in an exponential, or elliptic with one pole per period. For even  $p$ , however, Theorem 5.1 excludes other meromorphic waves only for purely dispersive equations.

Can one equation have two different waves of this kind at the same rate and speed? It cannot, even for the mixed equations of odd order, to which Theorem 5.1 does not apply.

**Proposition 7.5** (One profile per operator). *Let  $A_1, A_2$  be monic polynomials of degree  $p-1$  satisfying  $A_1 \mid S_{A_1}$  and  $A_2 \mid S_{A_2}$ . If  $P_{A_1} = P_{A_2}$ , then  $A_1 = A_2$ . Consequently a monic operator of degree  $p$  admits at most one normalized rational-exponential one-pole profile, and any count of distinct normalized pairs is also a count of distinct operators.*

*Proof.* Suppose  $P_{A_1} = P_{A_2} = P$  and  $h = A_1 - A_2 \neq 0$ . The degree  $n$  of  $h$  is less than  $p-1$ ; write  $f$  for its leading coefficient. By (3.3) and the bilinearity and symmetry of the convolution,  $S_{A_1} - S_{A_2} = h \star (A_1 + A_2) = (2/s_p)Ph$ . The power-sum calculation in (3.2) gives the leading coefficient  $2f \int_0^1 t^n (1-t)^{p-1} dt$  on the left. Since  $P$  is monic, the right side has the leading coefficient  $(2/s_p)f$ , and  $2/s_p$  is the beta integral  $\int_0^1 t^{p-1} (1-t)^{p-1} dt$  of (3.2). This is impossible:  $n < p-1$  implies  $t^n > t^{p-1}$  for  $0 < t < 1$ , so the first integral exceeds the second, and the left coefficient is more than twice the right one.  $\square$

Every  $P_A$  has a root among  $1, \dots, p$ . Let  $m$  be the least positive integer with  $A(m) \neq 0$ ; then  $m \leq p$ , because  $A$  has degree  $p-1$ . As  $e^z \rightarrow 0$  the profile begins with  $s_p(-1)^{m-1}A(m)e^{mz}$  and its square with  $e^{2mz}$ , so the coefficient of  $e^{mz}$  in (1.1) gives  $P_A(m) = 0$ . Physically, the wave leaves its background like  $e^{mz}$ , so  $m$  is a spatial eigenvalue of the linearization there, a root of  $P$ .

**Other equations with the same profile equation.** The reduction of Section 2 uses only the travelling-wave variable, so the counts apply to any equation  $\mathcal{L}u + \mathcal{M}(u^2) = 0$  with constant-coefficient operators, mixed derivatives and several space variables included, whose plane-wave reduction integrates to (1.1); the coefficients of  $P$  then depend on the speed and the direction of the wave. An example in three space variables is the equation of Kudryashov and Sinelshchikov [60] for a liquid with gas bubbles, whose reduction is that of KdV–Burgers. The repository lists such equations by profile operator (`tools/wave_dictionary/`).

**Ansatz methods.** In standard form (one phase  $kx - \omega t$ , constant coefficients) the tanh, sech, exp-function and  $(G'/G)$ -expansion methods and the simplest-equation method with a Riccati equation [46] return functions rational in one exponential, and the Jacobi and Weierstrass expansions [37] elliptic functions of one argument, with rational functions as limits. An output with one pole per period (for the equations of Theorem 5.1, every nonconstant output) is, up to the symmetries of Proposition 7.1, a wave (7.1), an elliptic wave of Section 4 or the rational wave. So the counts are a checklist: at a covered order, once the rate or the period lattice is fixed, a tanh-type or elliptic ansatz with one pole per period can return at most  $N_p$ , respectively  $M_p$ , normalized solutions, whatever the method. The critical comparisons [37, 42, 51] agree: the “new” KdV–Burgers and Kuramoto–Sivashinsky waves coincide with the known ones [42, 47], and an ansatz of pole order lower than  $p$  returns no wave [48]. The conformable derivative  $T_\alpha f(t) = t^{1-\alpha} f'(t)$ ,  $t > 0$ ,  $0 < \alpha \leq 1$  [30], is  $d/d\tau$  for  $\tau = t^\alpha/\alpha$ , and similarly in space, so a transformed equation with profile equation (1.1) obeys the same counts; this does not apply to the Caputo or Riemann–Liouville derivatives.

## 8 Examples

The examples show the counts of Table 1 at work and connect them with the classical lists. Order three was begun in Section 3.2.

### 8.1 Order two: KdV–Burgers

The equation  $u_t + uu_x + \nu u_{xx} + \delta u_{xxx} = 0$ ,  $\delta \neq 0$ , has  $p = 2$ ; the coefficient  $\nu$  may have either sign. It arises, for instance, for waves in a liquid with gas bubbles [53] and in a viscoelastic tube [44]. Its exact front solutions have been studied in several equivalent forms [27, 32, 37, 47, 73]. For  $A(\rho) = \rho + a$ , polynomial division gives

$$\text{rem}_A S_A = -\frac{a(a-1)(a+1)}{6}, \quad P_A = \rho^2 + 5a\rho + a^2 - 6a - 1, \quad v_A = 12((a+1)Q - Q^2). \quad (8.1)$$

Thus there are exactly  $N_2 = 3$  simple pairs, as Theorem 3.4 states for  $2p - 1 = 3$ :

| $a$ | $P_A(\rho)$            | $v_A$                                       |
|-----|------------------------|---------------------------------------------|
| −1  | $(\rho - 2)(\rho - 3)$ | $-12Q^2$                                    |
| 0   | $\rho^2 - 1$           | $12Q(1 - Q) = 3 \operatorname{sech}^2(z/2)$ |
| 1   | $(\rho - 1)(\rho + 6)$ | $12Q(2 - Q)$ .                              |

(8.2)

The pulse  $a = 0$  is the KdV soliton, the single pair of the purely dispersive subfamily,  $F_2 = 1$ . The two fronts belong to mixed equations and are mirror images of each other: for a given equation with  $\nu \neq 0$  they are one and the same front, read in the two directions (Proposition 7.1). In the original variables, (2.3) and (2.4) read

$$u(x, t) = w_* + \delta \kappa^2 v_A(\kappa(x - ct)), \quad \nu = \delta \kappa(5a), \quad w_* - c = \delta \kappa^2(a^2 - 6a - 1),$$

so a nonzero  $\nu$  fixes the front rate to  $\kappa = \nu/(5a\delta)$ ,  $a = \pm 1$ , in agreement with the classical parameter restriction. The elliptic pairs at this order, both for the KdV equation, are (4.4). Theorem 5.1 applies to the KdV subfamily  $a_1 = 0$ ; for the mixed operators the same conclusion cannot be expected (Remark 5.2).

## 8.2 Order three: elliptic pairs, and the lattices on which they merge

An elliptic solution must have zero residue at its pole. For  $P = D^3 + BD^2 + \mu D + a_0$  the Laurent residue is  $60\mu/19 - 15B^2/76$ , so  $B^2 = 16\mu$ ; this is the curve  $\sigma^2 = 16$  of (7.3). With this restriction the entire elliptic matching scheme is

$$\begin{aligned} P &= D^3 + BD^2 + \frac{B^2}{16}D + a_0, & v &= -60\wp' - 15B\wp - \frac{B^3}{64} - a_0, \\ B^4 &= 3072g_2, & a_0^2 &= \frac{13B^6}{2048} - 2160g_3. \end{aligned} \tag{8.3}$$

Read with the operator fixed, this is the classical solution of Kudryashov [34], see also Kudryashov [35]:  $w = v + a_0$  solves the usual integrated equation  $w''' + Bw'' + \mu w' + w^2/2 + K = 0$  with  $K = -a_0^2/2$ . For a given equation with  $B^2 = 16\mu$ ,  $g_2$  is fixed by  $B$  and  $g_3$  is free: a one-parameter family of elliptic waves. Hone [25] proves that the restriction  $B^2 = 16\mu$  is necessary, and Eremenko [23] completes the classification of all meromorphic KS solutions. For the count we read (8.3) the other way: fix  $g_2, g_3$  and solve for  $B$  and  $a_0$ .

On a generic lattice the equation for  $B$  has four roots and, for each of them, the equation for  $a_0$  has two, the two backgrounds  $\pm a_0$  of (2.5): these are the  $M_3 = 8$  pairs. On every lattice these two equations have eight solutions, counted with multiplicity. This is the count of Theorem 4.1: the unknowns eliminated on the way to (8.3) are determined linearly, with nonzero rational pivots, so its last two equations carry the full matching system, multiplicities included. On special lattices pairs merge, and the multiplicity of a merged pair is the number of pairs that have come together (Section 3). Pairs merge exactly where the Jacobian of the two equations vanishes, and that Jacobian is a nonzero constant times  $B^3 a_0$ . In terms of the invariant  $j(\Lambda)$  of Section 4: if  $B = 0$ , then  $g_2 = 0$ , hence  $j = 0$ ; if  $a_0 = 0$  on a lattice, elimination of  $B$  gives  $675g_3^2 = 169g_2^3$ , that is,  $j = -300$ . These conditions are sufficient as well. Thus:

| $j(\Lambda)$           | distinct pairs | total with multiplicity   |
|------------------------|----------------|---------------------------|
| $j \notin \{0, -300\}$ | 8              | $8 \times 1$              |
| -300                   | 6              | $2 \times 2 + 4 \times 1$ |
| 0                      | 2              | $2 \times 4$ .            |

(8.4)

This is the exceptional set of Theorem 4.1 at order three, and it shows the two ways in which pairs can meet. At  $j = -300$  the four values of  $B$  remain distinct, and for two of them the two backgrounds coincide at  $a_0 = 0$ , in one pair of multiplicity two; for instance,  $g_2 = 1/12$ ,  $g_3 = 13/1080$  gives the double points  $(B, a_0) = (\pm 4, 0)$ . At  $j = 0$  all four values of  $B$  meet at  $B = 0$ , and the two roots of  $a_0^2 = -2160g_3$  remain distinct, since  $g_3 \neq 0$ . Both pairs have the operator  $D^3 + a_0$ , which is purely dissipative.

## 8.3 Purely dispersive equations: Kawahara and seventh-order KdV

**Order four.** For the Kawahara equation [19, 28] the operator is

$$P = D^4 + a_2 D^2 + a_0. \tag{8.5}$$

The classification for this equation, using its first integral (5.5), is already established by Demina and Kudryashov [19]; see also Ye et al. [74]. By Theorem 6.2, the pairs with an even operator are

those with  $\iota A = A$ , that is,  $A = \rho^3 + \alpha_2 \rho$ . All of them are pulses: the profile is even about its pole,  $v(Q) = v(1 - Q)$ . The quotient and remainder are

$$\begin{aligned} P_A &= \rho^4 + 13\alpha_2 \rho^2 + \frac{31\alpha_2^2 - 70\alpha_2 + 7}{3}, \\ v_A &= 280(1 + \alpha_2)Q(1 - Q) - 1680Q^2(1 - Q)^2, \\ \text{rem}_A S_A &= -\frac{\rho}{420}(\alpha_2 + 1)(31\alpha_2^2 - 31\alpha_2 + 10). \end{aligned} \tag{8.6}$$

The three distinct roots,  $\alpha_2 = -1$  and  $\alpha_2 = 1/2 \pm 3i/(2\sqrt{31})$ , give  $F_4 = \binom{3}{1} = 3$  simple pairs. The cubic already appears in Kudryashov [33, Eq. (47)], with  $y = -1/\alpha_2$ , where one real and two complex conjugate roots are noted. The three values are the simply periodic parameter choices of Demina and Kudryashov [19, Section 3], whose parameter in Eq. (31) equals  $1/\alpha_2$ . At  $\alpha_2 = -1$ ,

$$v = -105 \operatorname{sech}^4(z/2), \quad P = (D^2 - 4)(D^2 - 9). \tag{8.7}$$

This is the familiar solitary wave of the Kawahara equation; the other two pairs are complex at unit real rate. Of all  $N_4 = 35$  pairs at this order, 21 have real coefficients, and all 21 have rational coefficients: five pulses and sixteen fronts, eleven wave shapes up to reflection. They are the one-pole waves of the mixed fifth-order equations, such as the Benney–Lin equation and the equation of Kudryashov and Sinelshchikov [54] for a viscoelastic tube, and the repository dictionary lists them; solitary waves of this family are found in Kudryashov and Demina [50].

Substitution shows that the elliptic pairs of (8.5) are given by

$$v = -1680\wp^2 - \frac{280a_2}{13}\wp + 168g_2 + \frac{31a_2^2}{507} - a_0, \tag{8.8}$$

$$0 = 31a_2^3 - 42588a_2g_2 - 4745520g_3, \tag{8.9}$$

$$a_0^2 = 23184g_2^2 + \frac{7980}{169}a_2^2g_2 + \frac{101520}{13}a_2g_3. \tag{8.10}$$

These are the Weierstrass profiles and parameter relations obtained from Laurent series by Demina and Kudryashov [19, Section 3]; in their notation  $a_2 = -\beta$ ,  $a_0^2 = C_0^2 + 12C_1$  and  $v = -6(w - w_*)$ , where their  $w_*$  is a constant equilibrium and  $a_0 = C_0 - 6w_*$ . The first condition is cubic in  $a_2$ , and the second is monic quadratic in  $a_0$ , so there are  $3 \times 2 = 6$  solutions, counted with multiplicity, on every lattice: three values of  $a_2$ , each with its two backgrounds. This is the number  $2F_4$  of Corollary 6.4. At  $(g_2, g_3) = (1, 0)$  the cubic has three distinct roots and the right side of (8.10) is nonzero at all of them, so the six pairs are simple on a nonempty open set of lattices. The unrestricted elliptic count at this order is  $M_4 = 30$ ; elliptic formulas with nonzero odd derivative coefficients in the full fifth-order evolution equation are given by Kudryashov and Sinelshchikov [59].

**Order six.** The seventh-order KdV equation has the operator  $P = D^6 + a_4D^4 + a_2D^2 + a_0$ ; its meromorphic travelling waves are studied by Dang [13]. Here  $A = \rho^5 + \alpha_2\rho^3 + \alpha_4\rho$  and  $s_6 = 5544$ . The remainder of  $S_A$  has two coefficients, at  $\rho^3$  and at  $\rho$ , which are polynomials in  $\alpha_2, \alpha_4$  of weights eight and ten. Eliminating  $\alpha_4$ , which is determined by  $\alpha_2$  at every solution, gives

$$(\alpha_2 + 5)(\alpha_2 + 10)(49\alpha_2^2 - 5\alpha_2 + 10)\varphi_6(\alpha_2) = 0, \tag{8.11}$$

where  $\varphi_6$  is an irreducible sextic without real roots; the ancillary script prints it and the two coefficients. The ten roots are distinct, in agreement with  $F_6 = \binom{5}{2} = 10$ , and all ten pairs are

pulses. Two pairs are real, the solitary waves of Ryabov et al. [71]:

$$\begin{array}{c|c|c}
A & P_A & v_A \\
\hline
\rho(\rho^2 - 1)(\rho^2 - 4) & (\rho^2 - 9)(\rho^2 - 16)(\rho^2 - 25) & 10395 \operatorname{sech}^6(z/2) \\
\rho(\rho^2 - 1)(\rho^2 - 9) & (\rho^2 - 4)(\rho^2 - 25)(\rho^2 - 71) & 10395 \operatorname{sech}^4(z/2)(1 + \operatorname{sech}^2(z/2)).
\end{array} \quad (8.12)$$

An exact computation on a sample lattice gives  $2F_6 = 20$  for the elliptic pairs of this family.

#### 8.4 Purely dissipative equations of order five

At  $p = 5$  the purely dissipative operators are  $P = D^5 + a_3 D^3 + a_1 D + a_0$ , and  $A = \rho^4 + \alpha_2 \rho^2 + \alpha_4$ . All 126 pairs at  $p = 5$  are simple (Appendix A). By Proposition 9.1(iii) there are therefore exactly  $F_5 = 6$  distinct pairs here, and all six are fronts,  $A(0) = \alpha_4 \neq 0$ ; for the four real ones this is also Theorem 6.2(iii). They are the three pairs in the first line of

$$\begin{aligned}
A = (\rho^2 - 1)(\rho^2 - 4), \quad A = (\rho^2 - 1)(\rho^2 - 9), \quad A = (\rho^2 - 1)\left(\rho^2 + \frac{1}{5}\right), \\
1105\alpha_2^3 - 2969\alpha_2^2 + 5570\alpha_2 - 4000 = 0, \quad \alpha_4 = \frac{5\alpha_2^3 + 21\alpha_2 - 10}{79\alpha_2 + 210},
\end{aligned} \quad (8.13)$$

and the three pairs given by the second line; the cubic has one real root. The six pairs are the six solitary waves of Kudryashov and Demina [49], in whose notation  $w = 1/(4a_3)$ . The four with real coefficients are, after the scales are restored, the solutions (1.4) of Kudryashov and Migita [41]; see also Ryabov et al. [71]. For instance,  $A = (\rho^2 - 1)(\rho^2 - 4)$  gives  $P_A = \rho^5 - 55\rho^3 + 1254\rho - 2520$  and  $v_A = 5040 Q^3(10 - 15Q + 6Q^2)$ , a front from 0 to  $5040 = -2P_A(0)$ .

### 9 Other orders and open questions

At a prime index  $\ell = 2p - 1$ , reduction modulo  $\ell$  gives rigidity and the simplicity of every pair. At the other orders these are two hypotheses, and the weighted Bézout bound replaces the labels (companion note `anc/general_orders.pdf` in the source package). The note proves the next proposition, Proposition 3.2, rigidity for prime powers  $2p - 1$  and the general reduction lemma behind Step 3 of the proof of Theorem 3.4, and it gives an algebraic proof of Lemma 4.2(i)–(ii).

**Proposition 9.1** (Every order). *Let  $p \geq 2$ .*

- (i) *The isolated solutions of the matching equations, counted with multiplicity, number at most  $N_p$  (Proposition 3.2), and those with  $\iota A = A$ , as solutions of the matching equations of the pure subfamily, at most  $F_p$  (Remark 6.3).*
- (ii) *If rigidity ( $R_{p-1}$ ) holds, all the systems below have finitely many solutions, and counted with multiplicity there are exactly  $N_p$  rational-exponential pairs (Proposition 3.2),  $S_p$  solutions of the pulse equations (3.18),  $F_p$  solutions of the matching equations of the pure subfamily,  $M_p$  elliptic pairs on every lattice and, for even  $p$ ,  $2F_p$  elliptic pairs with an even operator.*
- (iii) *If, in addition, every rational-exponential pair is simple, then all the conclusions of Theorems 3.4(a), 4.1 and 6.2, Proposition 3.6 and Corollary 6.4 hold at this  $p$ , except for the labelling.*

*Rigidity* ( $R_{p-1}$ ) is proved when  $2p - 1$  is a prime or a prime power. Exact integer certificates in the source package establish it for  $p \leq 73$  (degree  $d = p - 1 \leq 72$ ), and finite-field Gröbner computations in the repository (`certificates/rigidity_572/`), checked by independent pipelines, for  $p \leq 573$ . So the counts with multiplicity of part (ii) hold for  $2 \leq p \leq 573$ .

*Simplicity* of all pairs is established by exact computations at  $p = 5, 8$  and  $11$  (Appendix A). The other orders up to twelve have a prime index, so part (iii) applies for every  $p \leq 12$ ; the first order at which distinctness is not known is  $p = 13$ . Inside the two pure subfamilies a much smaller computation shows that the  $F_p$  pairs are distinct for every  $p \leq 14$ .

**Corollary 9.2** (Orders up to twelve). *For every  $2 \leq p \leq 12$  there are exactly  $N_p$  rational-exponential pairs, all simple, of which  $S_p$  are pulses and  $F_p$  belong to the pure subfamily of that order; and a generic lattice carries exactly  $M_p$  distinct elliptic pairs. In particular this holds for all the equations of Table 1.*

**Open questions.** The first is rigidity (3.8) in every degree; Lemma 3.3 says what it means for rational waves. The second is whether every point of  $\mathcal{X}_p$  is simple, that is, whether the tangent map of the matching equations at a solution  $S_A = AC_A$ ,

$$J_A(h) = \text{rem}_A(2(A \star h) - C_A h), \quad \deg h < d = p - 1, \quad (9.1)$$

always has zero kernel. At a prime index this holds already modulo  $\ell$ , where the map is multiplication by a polynomial coprime to  $\overline{A}$  (3.16); a uniform replacement is missing at composite indices. Simplicity does not follow from the uniqueness of the Laurent expansion in Theorem 5.1: that concerns a fixed operator, whereas in (9.1) the operator varies with the profile.

**Conjecture 9.3.** Rigidity (3.8) holds in every degree, and the matching Jacobian (9.1) is invertible at every point of  $\mathcal{X}_p$ . Consequently there are exactly  $\binom{2p-1}{p-1}$  rational-exponential pairs, all distinct, for every  $p \geq 2$ .

By Proposition 9.1(iii) the conjecture would extend Corollary 9.2 to every order, and the rational-exponential waves of purely dissipative equations would all be fronts. Simplicity on every lattice is false already at orders two (4.4) and three (8.4). For mixed equations of even order  $p \geq 4$  a separate question is open (Remark 5.2).

## A Exact verification and ancillary material

The five kinds of computer checks below use exact rational or finite-field arithmetic; instructions, dependency versions and checksums are included with the source package. The prime-index theorem does not depend on the computer calculations.

1. `verify_manuscript.py` reconstructs the convolutions, the first Laurent coefficients and the Weierstrass relations, and checks the leading-term comparison in Proposition 7.5 and, by exact substitution, the displayed formulas of orders two to four, including the agreement of Table 2 with the tables of Conte and Musette [7], Kudryashov and Zargaryan [43], the multiplicities (8.4), the Kawahara coefficient restriction and its literature normalization, the nodal correspondence and the Fisher identity. The uniform proofs are in the text.
2. `p5_count.sing` (SINGULAR, Gröbner basis over  $\mathbb{Q}$ ): at  $p = 5$  the count with multiplicity of the full matching equations, their quotient dimension, is 126. Adjoining the Jacobian determinant gives the unit ideal: the determinant vanishes at no solution, so all are simple.

3. At  $p = 8$  the same two outputs, 6435 and the unit ideal, are obtained over the finite field  $\mathbb{F}_{32003}$ . So every solution modulo 32003 is simple and lifts uniquely by Hensel’s lemma, as in Step 4 of the proof of Theorem 3.4, and the lifts are all the solutions because the count with multiplicity is 6435 (Proposition 9.1(ii)). At  $p = 11$  each of the 352716 solutions is certified over the 19-adic numbers, by Hensel’s lemma at the simple points modulo 19 and by exact Newton-polygon and Newton–Kantorovich certificates at the singular ones. The archived certificate and the record of its validation are in the repository (`certificates/`).
4. `rigidity/` has the proof and exact certificate for rigidity  $(R_d)$ ,  $d = p - 1 \leq 72$ . For  $2d + 1 = \ell^e + g$ ,  $g \in \{0, 2, 4, 6\}$ , the proof reduces the normalized continuous convolution to a residual system of degree at most  $g$ ; the prime-power case  $g = 0$  is also in the companion note. The certificate retains every residual degree, with integer ideal-membership identities and certified prime factors; the replay multiplies out every identity and checks all 72 arithmetic choices. It does not infer characteristic-zero nonexistence from lists of residue-field points.
5. `symmetry_checks/` checks Proposition 6.1 with symbolic coefficients for  $d \leq 5$ ; the labels of Table 2 and of the 21 real pairs at  $p = 4$ , with Proposition 3.6 and Theorem 6.2(ii) for them; the counts with multiplicity  $F_p$  for  $p \leq 8$ , and finite-field certificates, as at  $p = 8$ , that the  $F_p$  pairs are distinct for every  $p \leq 14$ ; on a sample lattice, the counts with multiplicity  $2F_p$  of Corollary 6.4 for  $p \leq 6$ , and no elliptic pair of a purely dissipative equation for  $p = 3, 5, 7$  (`elliptic_dissipative.py`); the six pairs (8.13) and the ten pairs (8.11), with the sextic factor in full; and the scaling (7.1) by substitution in the evolution equation.

The Lean 4 formalization of Remark 3.5 is at <https://github.com/migita/binomial-count-meromorphic-waves>, directory `lean/`, with Lean and Mathlib pinned to version 4.33.1. It proves rigidity  $(R_d)$  for  $2d + 1$  prime (`rigidity_prime_index`) and the count and simplicity statements of Theorem 3.4, both for the matching equations (3.5) (`primeIndexCountAndSimplicity`) and for the normalized pairs  $(P, v)$  as solutions of (1.1) in the field of rational functions of  $e^z$  (`primeIndexRationalExponentialCountAndSimplicity`), with the correspondence between the two formulations for every  $p \geq 2$ . The discrete convolution is defined there by the Bernoulli-polynomial power sums; the remainder equations and their Jacobian are those of Section 3. `EnumerationCheck.lean` and `Check.lean` verify that the proofs have exactly the stated propositions as their types and depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; the project contains no `sorry`, added axiom or `native_decide`.

## Tool and computational resource disclosure

This paper was written with substantial use of generative AI, which is disclosed here in the form recommended by the Leiden Declaration on Artificial Intelligence and Mathematics<sup>1</sup> and required by arXiv. No automated system is an author; credit and responsibility for the results, and for the attribution of prior work, rest with the author.

- **gpt-6-astra** (OpenAI), August–September 2026. The main mathematical ideas of the paper, the proofs of Theorems 5.1, 3.4 and 4.1 and of the weighted count in the companion note, the rigidity condition  $(R_d)$  and its certificate, the exact verification scripts, the SINGULAR input, the first complete draft of the text, and the Lean 4 formalization of Remark 3.5 were produced by this model in dialogue with the author.

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<sup>1</sup><https://leidendeclaration.ai/>, 2 June 2026.

- OpenAI Codex v0.153.4, September 2026. Independent mathematical audit of the final revision, integration of the edits, and rebuilding and validation of the source package.
- Claude Code (Anthropic, Claude Fable 5.1), September 2026. The literature search behind the attributions in Section 1, the readability revision of all sections, and the leading-coefficient argument of Proposition 7.5, which Codex then checked independently. In the final revision it proposed and proved the results on the two pure subfamilies and on single equations (Corollary 5.3, Section 6, Corollary 6.4 and Section 7), with their exact checks, and reorganized the manuscript around them. In the last revision it made the prime-index theorem the spine of the paper: it wrote the proof of Theorem 3.4 in the form used by the Lean formalization, derived Proposition 3.6 and Theorem 6.2 from the labels, replaced the Bézout count in the proof of Theorem 4.1 by a flat-family argument, and moved the general-order material to the companion note; Codex reviewed these arguments. It also audited the Lean formalization independently: the formal statements against the manuscript, a rebuild from source, and the axiom audit. In the present revision it reorganized the exposition for readers in nonlinear science: the counts now precede the classification, the proof of Theorem 3.4 is presented through a dominant-balance argument and Hensel’s lemma, and the proof of Theorem 4.1 through a conservation-of-number lemma, the flat-family argument having moved to the companion note.

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